

State-independent projector-sum identities: Real–complex thresholds and hypergraph forcing

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(Dated: July 10, 2026)*

We study *pseudocontexts*: disjoint ray families with equal sums of rank-one projectors, and hence equal Born-weight sums in every quantum state. Genuine examples exist for every $k \geq 2$ over $\mathbb{C}$, but first occur at $k = 3$ over $\mathbb{R}$; the $k = 2$ dichotomy is due to complex relative phases. We distinguish representation-dependent identities from identities forced by orthogonality-hypergraph context equations. A 20-ray octagon gives a graph-forced size-two identity and an algebraic $\mathbb{C}^3$ realization; its 21-ray $C = 1$ augmentation has no faithful real $\mathbb{R}^3$ realization when each triple is a basis. Tight frames provide full-support real families.

I. INTRODUCTION

Let L be the lattice of projections in a finite-dimensional real or complex Hilbert space. Its atoms are one-dimensional subspaces, or *rays*. The state convention is important: throughout this paper, unless explicitly stated otherwise, a state means a density-operator (Born-rule) state. Thus the probability of a ray a is $p_\rho(a) = \text{tr}(\rho P_a)$. In dimension at least three this agrees with the usual lattice-state description by Gleason’s theorem; in dimension two it is a deliberate restriction to quantum states.

Vectors $a_1, \dots, a_n$ form a *context* if they form an orthonormal basis. In this case (and only in this case) they satisfy

$$\forall \rho : \sum_{i=1}^n p_\rho(a_i) = 1. \quad (1)$$

We investigate weaker additive identities between nonorthogonal ray families. They are naturally frame-theoretic: equality of the two sums in every quantum state is exactly equality of two rank-one projector sums. The resulting identities should not be conflated with contextuality witnesses: they can be representation-dependent, or they can follow from the linear context equations of a hypergraph, and either kind can coexist with classical two-valued states. The purpose of this article is to separate these mechanisms and to identify the real–complex threshold for genuinely noncontextual ray families.

Table I records the distinctions used throughout. Our new contributions are the sharp size threshold, the octagon covering and coordinatizations, and the real–complex representability obstruction. The frame, Gram,

Schur–Horn, and commutant facts collected later are standard tools used to organize these results.

The paper is organized as follows. Sections II–V establish the threshold and introduce the necessary frame tools. Sections VIII–IX distinguish hypergraph forcing from representation dependence and analyze the octagon. Section XI gives full-support real constructions, and Section XIII summarizes the scope and limitations.

II. PRELIMINARIES

Throughout, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, H is an n -dimensional Hilbert space over $\mathbb{F}$ with $n \geq 2$, and $L = L(H)$ is the orthomodular lattice of its closed subspaces. Its atoms correspond to one-dimensional subspaces, or equivalently to *rays* spanned by nonzero vectors. We identify an atom with any representative unit vector a and write $P_a = |a\rangle\langle a|$ for the orthogonal projection onto the ray it spans. More generally, for nonzero a we set $P_a = |a\rangle\langle a|/\langle a, a\rangle$, so that P_a depends only on the ray $\mathbb{F}a$.

The quantum state space used in this paper is

$$p_\rho(a) = \text{tr}(\rho P_a), \quad \rho \geq 0, \quad \text{tr} \rho = 1. \quad (2)$$

We denote this set by $S_q(L)$ and, for compactness, write $S(L) = S_q(L)$ below. In dimension $n \geq 3$, Gleason’s theorem [1] identifies these with the usual finitely additive lattice states. In $n = 2$, non-Born lattice states exist; the results in this paper concern $S_q(L)$, not that larger abstract state space. The key consequence used repeatedly is the following elementary separation fact.

Lemma 1 (Separation by density operators). *Let A and B be self-adjoint operators on a finite-dimensional real or complex Hilbert space. If*

$$\text{tr}(\rho A) = \text{tr}(\rho B)$$

for every density operator ρ , then $A = B$.

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TABLE I. Taxonomy of the principal examples. “Bare graph” refers only to the orthogonality hypergraph induced by the displayed rays; it does not rule out a forcing auxiliary extension.

Example	Classification
Complex $k = 2$ pair	genuine; non-scalar T ; $\mathbb{C}^2$; representation-dependent, and the bare graph does not force it
Mercedes–Benz pair	genuine; scalar T on $\mathbb{R}^2$; operator-level identity
$C \neq 1$ octagon	genuine; non-scalar T ; algebraic $\mathbb{C}^3$ representation; graph-forced by a covering certificate
$C = 1$ augmentation	nongenuine; scalar T on $c^\perp$; $\mathbb{C}^3$ realization; graph-forced, with inner pairs relative contexts
Tetrahedral pair	genuine; scalar T on $\mathbb{R}^3$; representation-dependent; finite forcing extension open

Proof. Apply the hypothesis to $C := A - B$. For every unit vector x , the rank-one projection P_x is a density operator, hence

$$0 = \text{tr}(P_x C) = \langle x, Cx \rangle.$$

By homogeneity the quadratic form $q(x) = \langle x, Cx \rangle$ vanishes for every vector x . Since $C = A - B$ is self-adjoint, the associated sesquilinear form is determined by this quadratic form. In the real case,

$$\langle x, Cy \rangle = \frac{1}{4}(q(x+y) - q(x-y)) = 0,$$

and in the complex case, with the convention that the inner product is linear in the second argument,

$$\langle x, Cy \rangle = \frac{1}{4} \sum_{\ell=0}^3 (-i)^\ell q(x + i^\ell y) = 0.$$

Thus $\langle x, Cy \rangle = 0$ for all x, y , hence $C = 0$. $\square$

III. CONTEXTS AND PSEUDOCONTEXTS

The operator form of (1) is elementary and is the starting point for everything that follows.

Proposition 2 (Contexts). *Unit vectors $a_1, \dots, a_n \in H$ form an orthonormal basis if and only if*

$$\sum_{i=1}^n P_{a_i} = I. \quad (3)$$

Proof. The forward implication is immediate. For the converse, fix j and evaluate (3) on the vector a_j :

$$1 = \langle a_j, I a_j \rangle = \sum_{i=1}^n |\langle a_i, a_j \rangle|^2.$$

The term with $i = j$ already contributes 1, so all others must vanish. Hence $\langle a_i, a_j \rangle = 0$ for $i \neq j$. As this holds for every j , the family is orthonormal, and being of size n in an n -dimensional space it is an orthonormal basis. $\square$

Combined with (2), Proposition 2 recovers (1) as a state-theoretic characterisation of orthogonal bases.

Definition 3 (Pseudocontext). Let $k \geq 1$. A *pseudocontext of size k* is a pair of ordered representatives of two k -element ray families $(a_1, \dots, a_k), (b_1, \dots, b_k)$ such that

- (i) each tuple contains pairwise distinct atoms (i.e. no two of the a_i are scalar multiples, and likewise for the b_i),
- (ii) the two tuples are disjoint as sets of rays, so that $|\{a_1, \dots, a_k, b_1, \dots, b_k\}| = 2k$, and

$$(iii) \quad \forall \rho : \sum_{i=1}^k p_\rho(a_i) = \sum_{i=1}^k p_\rho(b_i).$$

A pseudocontext is called *genuine* if neither k -tuple is a context on its supporting subspace, i.e. if the rays in neither tuple form an orthonormal basis of their span.

Remark 4. Condition (i) excludes repeated atoms within a single tuple, and condition (ii) ensures the two tuples are truly distinct as collections of rays. Without condition (ii), every pseudocontext of size k would yield pseudocontexts of every size $k' > k$ by adjoining $k' - k$ shared atoms; this trivial inflation can be carried out within the original ambient subspace. Condition (ii) rules out such padding and makes the size k a genuine invariant of the construction.

Remark 5 (Contexts on supporting subspaces). We call an orthonormal basis of the supporting subspace a *relative context*. Thus, whenever “context on the support” is used below, it means a relative context; an ambient context is an orthonormal basis of the full Hilbert space.

By (2) and Lemma 1, condition (iii) of Definition 3 is equivalent to the operator identity

$$T := \sum_{i=1}^k P_{a_i} = \sum_{i=1}^k P_{b_i}, \quad (4)$$

which we call the *structure operator* of the pseudocontext.

Proposition 6 (Genuineness criterion). *Let $(a_1, \dots, a_k), (b_1, \dots, b_k)$ be a pseudocontext with structure operator*

$$T = \sum_{i=1}^k P_{a_i} = \sum_{i=1}^k P_{b_i}.$$

Then the following are equivalent:

1. the pseudocontext is genuine;
2. T is not an orthogonal projection onto its range;
3. $\lambda_{\max}(T) > 1$.

Proof. The range of T is $\text{span}\{a_1, \dots, a_k\}$, equivalently $\text{span}\{b_1, \dots, b_k\}$. The a -tuple is a context on its supporting subspace if and only if

$$T = P_{\text{ran } T}.$$

Indeed, this is exactly Proposition 2 applied inside the supporting subspace. If this holds for the a -tuple, then the same operator identity holds for the b -tuple, so the b -tuple is also a context on the same support. Hence non-genuineness is equivalent to T being an orthogonal projection.

It remains to relate this to $\lambda_{\max}(T)$. Since T is a sum of k rank-one projections, $\text{rank } T \leq k$ and $\text{tr } T = k$. If $\lambda_{\max}(T) \leq 1$, then

$$k = \text{tr } T \leq \text{rank } T \leq k,$$

so equality holds throughout and all nonzero eigenvalues of T are equal to 1. Thus T is a projection. Conversely, if T is a projection, then $\lambda_{\max}(T) = 1$. Therefore T is not a projection if and only if $\lambda_{\max}(T) > 1$. $\square$

Thus pseudocontexts are precisely pairs of distinct unit-norm rank-one decompositions of a single positive operator. This places them naturally among finite frame-theoretic constructions: contexts are the special case in which the structure operator is the identity on the relevant subspace; in general, pseudocontexts are unit-norm rank-one frame decompositions of a common positive operator. If the common operator is $(k/r)I$ on a support of dimension r , the effects $(r/k)P_{a_i}$ form a rank-one POVM on that support.

IV. THE SMALLEST CASES

We begin with the minimum-size question: size $k = 1$ is impossible, size $k = 2$ exists over $\mathbb{C}$, and size $k = 2$ has no genuine real realisation.

A. No pseudocontexts of size $k = 1$

Proposition 7 (Size $k = 1$). *There is no pseudocontext of size $k = 1$.*

Proof. The condition $P(a_1) = P(b_1)$ for all $P \in S(L)$ means that the two self-adjoint operators P_{a_1} and P_{b_1} have equal expectation in every density operator. By Lemma 1, $P_{a_1} = P_{b_1}$. This rank-one identity forces a_1 and b_1 to span the same ray, contradicting the distinctness condition (ii) of Definition 3. $\square$

B. Pseudocontexts of size $k = 2$ and the Bloch representation

In dimension two the pseudocontext condition admits a complete characterisation in terms of Bloch vectors. This yields a unified and basis-independent description of all pseudocontexts and clarifies the real-complex dichotomy.

Every rank-one projector on $\mathbb{C}^2$ can be written as

$$P_a = \frac{1}{2}(I + \vec{r}_a \cdot \vec{\sigma}), \quad \vec{r}_a \in \mathbb{R}^3, \quad \|\vec{r}_a\| = 1, \quad (5)$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices. To be specific, for any unit vector $a = \alpha e_1 + \beta e_2 \in \mathbb{C}^2$ in a chosen orthonormal basis (e_1, e_2) , the components of the associated Bloch vector $\vec{r}_a = (r_x, r_y, r_z)$ are obtained by computing the expectation values of the Pauli matrices:

$$\begin{aligned} r_x &= \langle a, \sigma_x a \rangle = 2 \text{Re}(\alpha^* \beta), \\ r_y &= \langle a, \sigma_y a \rangle = 2 \text{Im}(\alpha^* \beta), \\ r_z &= \langle a, \sigma_z a \rangle = |\alpha|^2 - |\beta|^2. \end{aligned} \quad (6)$$

Proposition 8 (Bloch representation of pseudocontexts in dimension $d = 2$). *Let $(a_1, \dots, a_k)$ and $(b_1, \dots, b_k)$ be two tuples of unit vectors in $\mathbb{C}^2$ satisfying the distinctness/disjointness conditions of Definition 3(i)–(ii). Then the following are equivalent:*

1. condition (iii) of Definition 3 holds;

$$2. \sum_{i=1}^k P_{a_i} = \sum_{i=1}^k P_{b_i};$$

$$3. \sum_{i=1}^k \vec{r}_{a_i} = \sum_{i=1}^k \vec{r}_{b_i}.$$

The common structure operator T has Bloch form

$$T = \frac{k}{2}I + \frac{1}{2} \vec{R} \cdot \vec{\sigma}, \quad \vec{R} = \sum_{i=1}^k \vec{r}_{a_i} = \sum_{i=1}^k \vec{r}_{b_i}.$$

Proof. By Definition 3 and (4), conditions 1 and 2 are equivalent. For the equivalence with 3, substitute the Bloch representation (5) into the structure operator:

$$\sum_{i=1}^k P_{a_i} = \sum_{i=1}^k \frac{1}{2}(I + \vec{r}_{a_i} \cdot \vec{\sigma}) = \frac{k}{2}I + \frac{1}{2} \left(\sum_{i=1}^k \vec{r}_{a_i} \right) \cdot \vec{\sigma}.$$

Since $I, \sigma_x, \sigma_y, \sigma_z$ are linearly independent, equality of the two structure operators is equivalent to equality of the Bloch-vector sums. $\square$

Remark 9. When such a two-dimensional construction is embedded into a larger Hilbert space, the Bloch representation refers to the two-dimensional supporting subspace; the structure operator is extended by zero on its orthogonal complement.

C. Explicit $k = 2$ example via Bloch vectors

We can now construct an explicit $k = 2$ pseudocontext by directly applying the Bloch representation and showing exactly how it arises from non-orthogonal configurations.

Proposition 10 (Existence of $k = 2$ complex pseudocontexts). *There are genuinely nonorthogonal pseudocontexts of size $k = 2$ in $\mathbb{C}^n$ for every $n \geq 2$.*

Proof. Embed $\mathbb{C}^2$ into $\mathbb{C}^n$ as the span of the first two basis vectors e_1, e_2 . Define two pairs of unit vectors:

$$u_{\pm} = \left(\frac{\sqrt{3}}{2}, \pm \frac{1}{2}, 0, \dots, 0 \right), \quad v_{\pm} = \left(\frac{\sqrt{3}}{2}, \pm \frac{i}{2}, 0, \dots, 0 \right).$$

We obtain their Bloch vectors explicitly using (6). For $u_{\pm}$, setting $\alpha = \sqrt{3}/2$ and $\beta = \pm 1/2$, we compute:

$$\begin{aligned} r_x &= 2 \operatorname{Re}\left(\frac{\sqrt{3}}{2} \cdot (\pm \frac{1}{2})\right) = \pm \frac{\sqrt{3}}{2}, \\ r_y &= 2 \operatorname{Im}\left(\frac{\sqrt{3}}{2} \cdot (\pm \frac{1}{2})\right) = 0, \\ r_z &= \left|\frac{\sqrt{3}}{2}\right|^2 - \left|\pm \frac{1}{2}\right|^2 = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}. \end{aligned}$$

Thus $\vec{r}_{u_{\pm}} = (\pm \frac{\sqrt{3}}{2}, 0, \frac{1}{2})$, which lie in the xz -plane of the Bloch sphere.

For $v_{\pm}$, setting $\alpha = \sqrt{3}/2$ and $\beta = \pm i/2$, we have $\alpha^* \beta = (\sqrt{3}/2)(\pm i/2)$, so:

$$\begin{aligned} r_x &= 2 \operatorname{Re}\left(\frac{\sqrt{3}}{2} \cdot \pm \frac{i}{2}\right) = 0, \\ r_y &= 2 \operatorname{Im}\left(\frac{\sqrt{3}}{2} \cdot \pm \frac{i}{2}\right) = \pm \frac{\sqrt{3}}{2}, \\ r_z &= \frac{3}{4} - \frac{1}{4} = \frac{1}{2}. \end{aligned}$$

Thus $\vec{r}_{v_{\pm}} = (0, \pm \frac{\sqrt{3}}{2}, \frac{1}{2})$, which lie in the yz -plane.

Summing the Bloch vectors within each pair yields the same structure vector:

$$\vec{R} = \vec{r}_{u_+} + \vec{r}_{u_-} = \vec{r}_{v_+} + \vec{r}_{v_-} = (0, 0, 1).$$

By Proposition 8, this guarantees that the operator sums match in the two-dimensional subspace. Embedding back into $\mathbb{C}^n$ gives, with $V := \operatorname{span}\{e_1, e_2\}$,

$$\begin{aligned} P_{u_+} + P_{u_-} &= P_{v_+} + P_{v_-} = \frac{3}{2}P_{e_1} + \frac{1}{2}P_{e_2} \\ &= (I_V + \frac{1}{2}\sigma_z) \oplus 0_{n-2} = \begin{pmatrix} 3/2 & 0 \\ 0 & 1/2 \end{pmatrix} \oplus 0_{n-2}. \end{aligned}$$

The four rays are distinct: u_+ and u_- differ by a sign in the real second coordinate; v_+ and v_- differ by a sign in the imaginary second coordinate. No $u_{\pm}$ is collinear with any $v_{\pm}$: comparison of the first coordinate forces the proportionality scalar to be 1, but then the second coordinates are different. Within each pair, $\langle u_+, u_- \rangle = \frac{1}{2} \neq 0$ and $\langle v_+, v_- \rangle = \frac{1}{2} \neq 0$, so neither pair is orthogonal. The pseudocontext is therefore genuine. $\square$

Remark 11 (Bloch geometry of the complex $k = 2$ example). This construction directly illustrates the continuous $U(1)$ freedom present in complex spaces. The pair of Bloch vectors $\{\vec{r}_{v_+}, \vec{r}_{v_-}\}$ is related to $\{\vec{r}_{u_+}, \vec{r}_{u_-}\}$ by a 90° rotation about the z -axis (which aligns with their common sum $\vec{R}$). Because this rotation preserves $\vec{R}$, it produces a distinct, valid decomposition of the same fixed structure operator T .

D. Bloch-sphere description in $\mathbb{C}^2$

Theorem 12 (Bloch-sphere characterisation in $\mathbb{C}^2$). *Under the Bloch map, pseudocontexts of size k in $\mathbb{C}^2$ are exactly pairs of disjoint k -point subsets of the unit sphere $S^2 \subset \mathbb{R}^3$ having the same vector sum. Equivalently,*

$$\sum_{i=1}^k P_{a_i} = \sum_{i=1}^k P_{b_i} \iff \sum_{i=1}^k \vec{r}_{a_i} = \sum_{i=1}^k \vec{r}_{b_i}.$$

Consequently:

- (i) For $k = 1$, no pseudocontext exists.
- (ii) For $k = 2$, pseudocontexts exist and form continuous families.
- (iii) For every $k \geq 3$, genuine examples exist. In particular, zero-centroid examples are obtained by taking two disjoint regular k -gons on a common great circle of the Bloch sphere.

Proof. Part (i) is Proposition 7.

For (ii), fix $\vec{R} \in \mathbb{R}^3$ with $0 < \|\vec{R}\| < 2$. Writing

$$\vec{r}_1 = \vec{R}/2 + \vec{w}, \quad \vec{r}_2 = \vec{R}/2 - \vec{w},$$

the condition $\|\vec{r}_1\| = \|\vec{r}_2\| = 1$ is equivalent to $\vec{w} \perp \vec{R}$ and $\|\vec{w}\|^2 = 1 - \|\vec{R}\|^2/4$. Thus $\vec{w}$ ranges over a circle in the plane orthogonal to $\vec{R}$, so the set of unordered pairs $\{\vec{r}_1, \vec{r}_2\}$ is a continuous family.

For (iii), choose any $k \geq 3$ and take a regular k -gon on a great circle of the Bloch sphere. Let $(a_1, \dots, a_k)$ be its corresponding rays in $\mathbb{C}^2$. Let $(b_1, \dots, b_k)$ be obtained by rotating the polygon in the same great circle through an angle δ which is not a multiple of $2\pi/k$. Then no rotated vertex coincides with an original vertex, so the two ray sets are disjoint. Both k -tuples have centroid 0, hence the same structure operator, and because $k \geq 3$ neither tuple is a context. Therefore they form a genuine pseudocontext. $\square$

E. Real-complex dichotomy in two dimensions

The $k = 2$ complex example in Proposition 10 is the simplest continuous family. In $\mathbb{R}^2$, no genuine pseudocontext of size $k = 2$ exists, as stated in Theorem 13(ii) below.

Theorem 13 (Real vs. complex in $d = 2$).

- (i) In $\mathbb{C}^2$, genuine pseudocontexts of size $k = 2$ exist.
- (ii) In $\mathbb{R}^2$, no genuine pseudocontext of size $k = 2$ exists.
- (iii) In $\mathbb{R}^2$, genuine pseudocontexts exist for all $k \geq 3$.

Proof. Part (i) follows from Proposition 10.

For (ii), represent a real ray by a unit vector $x_\theta = (\cos \theta, \sin \theta)$, with angles understood modulo π . Its projector is

$$P_\theta = \frac{1}{2} \begin{pmatrix} 1 + \cos 2\theta & \sin 2\theta \\ \sin 2\theta & 1 - \cos 2\theta \end{pmatrix}.$$

Thus a real ray is parametrised by the doubled-angle unit vector $q_\theta = (\cos 2\theta, \sin 2\theta)$, and equality $P_{\theta_1} + P_{\theta_2} = P_{\phi_1} + P_{\phi_2}$ is equivalent to $q_{\theta_1} + q_{\theta_2} = q_{\phi_1} + q_{\phi_2}$. If this common sum is nonzero, a two-term decomposition of it into unit vectors in the plane is unique up to permutation, so the two unordered pairs of rays coincide. If the common sum is zero, then $q_{\theta_2} = -q_{\theta_1}$, hence the two corresponding rays are orthogonal; the pair is a context on its two-dimensional support. In either case no genuine real pseudocontext of size 2 exists.

For (iii), choose distinct real rays with angles

$$\theta_j = \theta_0 + \frac{\pi j}{k}, \quad j = 0, \dots, k-1.$$

Their doubled-angle vectors form a regular k -gon and therefore have centroid 0. A second copy rotated by an angle $\delta \not\equiv m\pi/k \pmod{\pi}$ for all $m \in \mathbb{Z}$ is disjoint from the first one and has the same centroid. The two projector sums are consequently equal, and since $k \geq 3$ neither tuple is an orthonormal basis of its two-dimensional support. Hence the two tuples form a genuine pseudocontext. $\square$

Remark 14 (Earlier observation). The nonexistence of genuine real pseudocontexts of size $k = 2$ in dimension two was observed in [2]. The doubled-angle (real Bloch-circle) parametrisation makes the real-complex dichotomy transparent: over $\mathbb{R}$ a non-scalar two-term decomposition is unique up to permutation, whereas over $\mathbb{C}$ a continuous phase degree of freedom produces inequivalent decompositions. The scalar case $T = I$ instead has many orthonormal-basis decompositions, all nongenuine.

Proposition 15. *No genuine pseudocontext of size $k = 2$ exists over a real Hilbert space of any finite dimension.*

Proof. Suppose

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2} =: T$$

with all vectors real and unit. The range of T is

$$\text{span}\{a_1, a_2\} = \text{span}\{b_1, b_2\},$$

so all four rays lie in a subspace of dimension at most two. If this subspace is one-dimensional, distinctness fails. Hence we may restrict to a real two-dimensional support.

If $a_1 \perp a_2$, then T acts as the identity on its two-dimensional support. Proposition 2, applied within that support, then implies that any two-vector unit decomposition of the identity is an orthonormal basis of the support; so both pairs are contexts and the pseudocontext is not genuine.

If a_1 and a_2 are not orthogonal, choose their signs so that $\gamma = \langle a_1, a_2 \rangle \in (0, 1)$. Then T has the simple eigenvalues $1 + \gamma$ and $1 - \gamma$, with eigenvectors proportional to $a_1 + a_2$ and $a_1 - a_2$.

We spell out the uniqueness. Suppose also

$$T = P_{c_1} + P_{c_2}$$

with real unit vectors c_1, c_2 . Choosing the sign of c_2 , assume $\delta = \langle c_1, c_2 \rangle \geq 0$. The nonzero eigenvalues of $P_{c_1} + P_{c_2}$ are $1 + \delta$ and $1 - \delta$. Since the eigenvalues of T are $1 + \gamma$ and $1 - \gamma$, we have $\delta = \gamma$. The corresponding eigenvectors are proportional to $c_1 + c_2$ and $c_1 - c_2$, hence, by simplicity of the spectrum,

$$c_1 + c_2 = \lambda(a_1 + a_2), \quad c_1 - c_2 = \mu(a_1 - a_2)$$

for some real nonzero λ, μ . Therefore

$$c_1 = \frac{1}{2}(\lambda(a_1 + a_2) + \mu(a_1 - a_2)).$$

Using $\langle a_1 + a_2, a_1 - a_2 \rangle = 0$, the conditions $\|c_1\| = 1$ and $\langle c_1, c_2 \rangle = \gamma$ give

$$\lambda^2(1 + \gamma) + \mu^2(1 - \gamma) = 2,$$

and

$$\lambda^2(1 + \gamma) - \mu^2(1 - \gamma) = 2\gamma.$$

Solving yields $\lambda^2 = \mu^2 = 1$. Since $\lambda, \mu \in \mathbb{R}$, this gives only the sign choices $\lambda, \mu = \pm 1$, which recover the same two rays $\{[a_1], [a_2]\}$. Thus no disjoint second real decomposition exists in the nonorthogonal case. $\square$

Remark 16. The same calculation over $\mathbb{C}$ gives $|\lambda| = |\mu| = 1$ instead of $\lambda, \mu = \pm 1$. The resulting relative phase is precisely the continuous freedom responsible for the complex $k = 2$ pseudocontexts of Proposition 10.

F. Spectral form

Proposition 17 (Spectrum in $d = 2$). *The eigenvalues of T are*

$$\lambda_{\pm} = \frac{k}{2} \pm \frac{1}{2} \|\vec{R}\|.$$

Proof. By Proposition 8,

$$T = \frac{k}{2}I + \frac{1}{2}\vec{R} \cdot \vec{\sigma}.$$

The Pauli identity

$$(\vec{R} \cdot \vec{\sigma})^2 = \|\vec{R}\|^2 I$$

implies that $\vec{R} \cdot \vec{\sigma}$ has eigenvalues $\pm \|\vec{R}\|$. Hence T has eigenvalues

$$\lambda_{\pm} = \frac{k}{2} \pm \frac{1}{2}\|\vec{R}\|.$$

□

Remark 18. The vector $\vec{R}$ completely parametrises the structure operator in dimension two. It does not parametrise a pseudocontext: a fixed structure operator can have continuously many inequivalent decompositions.

V. PSEUDOCONTEXTS OF EVERY SIZE $k \geq 2$

The $k = 2$ case is best understood via the Bloch-sphere parametrisation; the general k -construction below is its root-of-unity extension. This section supplies the complex existence statement announced in the Introduction for all $k \geq 2$. The construction below uses a continuous phase $e^{i\phi}$ and is therefore intrinsically complex; the corresponding existence statement over $\mathbb{R}$ for $k \geq 3$ is covered separately by Theorem 13(iii); Example 22 gives the smallest case explicitly.

Theorem 19 (Complex-phase construction). *Let H contain an orthonormal pair e_1, e_2 . Fix an integer $k \geq 2$, let $\omega = e^{2\pi i/k}$, choose $\alpha \in (0, \pi/2)$, and choose $\phi \in \mathbb{R}$ such that $e^{i\phi}$ is not a k th root of unity. Define unit vectors*

$$\begin{aligned} a_j &= \cos \alpha e_1 + \sin \alpha \omega^j e_2, \\ b_j &= \cos \alpha e_1 + \sin \alpha e^{i\phi} \omega^j e_2, \quad j = 0, \dots, k-1. \end{aligned}$$

Then the $2k$ rays spanned by the a_j and b_j are pairwise distinct, and

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k(\cos^2 \alpha P_{e_1} + \sin^2 \alpha P_{e_2}).$$

Hence $(a_0, \dots, a_{k-1})$ and $(b_0, \dots, b_{k-1})$ form a pseudocontext of size k . The pseudocontext is genuine whenever $k \geq 3$ or $\alpha \neq \pi/4$.

Proof. Write $c = \cos \alpha$ and $s = \sin \alpha$. For each j ,

$$\begin{aligned} P_{a_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs(\omega^{-j}|e_1\rangle\langle e_2| + \omega^j|e_2\rangle\langle e_1|), \\ P_{b_j} &= c^2 P_{e_1} + s^2 P_{e_2} + cs(e^{-i\phi}\omega^{-j}|e_1\rangle\langle e_2| + e^{i\phi}\omega^j|e_2\rangle\langle e_1|). \end{aligned}$$

Summing over $j = 0, \dots, k-1$ and using $\sum_{j=0}^{k-1} \omega^j = 0$ cancels all off-diagonal terms, giving

$$\sum_{j=0}^{k-1} P_{a_j} = \sum_{j=0}^{k-1} P_{b_j} = k(c^2 P_{e_1} + s^2 P_{e_2}).$$

Distinctness. If $a_j = \lambda a_\ell$ for some scalar λ , then comparing first coordinates (which equal $c \neq 0$) gives $\lambda = 1$, and then $\omega^j = \omega^\ell$, so $j = \ell$. Hence the a_j are pairwise distinct rays; the identical argument applies to the b_j . If $a_j = \lambda b_\ell$, then $\lambda = 1$ and $\omega^j = e^{i\phi}\omega^\ell$, i.e. $e^{i\phi} = \omega^{j-\ell}$, contradicting the assumption that $e^{i\phi}$ is not a k th root of unity. Hence no a_j coincides with any b_ℓ .

Genuineness. For $j \neq \ell$ within the a -tuple,

$$\langle a_j, a_\ell \rangle = c^2 + s^2 \omega^{\ell-j}.$$

This vanishes only if $c^2 = s^2$ (i.e. $\alpha = \pi/4$) and $\omega^{\ell-j} = -1$. Thus, if $\alpha \neq \pi/4$, no two distinct vectors in the a -tuple are orthogonal. An identical computation gives $\langle b_j, b_\ell \rangle = c^2 + s^2 \omega^{\ell-j}$, so the same conclusion holds for the b -tuple. Hence neither tuple is a context when $\alpha \neq \pi/4$. If instead $k \geq 3$, each tuple consists of k distinct rays inside the two-dimensional support $\text{span}\{e_1, e_2\}$, and therefore cannot be an orthonormal basis of that support. Hence the pseudocontext is genuine whenever $k \geq 3$ or $\alpha \neq \pi/4$. □

Remark 20 (Complex versus real constructions). Theorem 19 uses a genuinely complex phase. The real examples in Example 22 and Section XI are different mechanisms and should not be conflated with the root-of-unity family.

Remark 21. At the special value $\alpha = \pi/4$, the structure operator becomes $\frac{k}{2}I_{\text{span}\{e_1, e_2\}}$, so the construction yields a unit-norm tight frame on the two-dimensional support. For $k = 2$ this reduces to a context; for $k \geq 3$ it gives a highly symmetric pseudocontext (cf. Remark 23). In dimension $d = 2$, this construction reduces to the Bloch-vector description of Sec. IV B.

VI. A SYMMETRIC $k = 3$ EXAMPLE

The case $k = 3$ admits a very transparent planar representation; it is the finite-dimensional analogue of the Mercedes-Benz frame.

Example 22 (Mercedes-Benz type pseudocontext). Fix a two-dimensional subspace $V \subset H$ with orthonormal basis (e_1, e_2) , and fix an angle θ satisfying $\theta \neq 0$, $\theta \neq \pi/3$, and $\theta \neq 2\pi/3$ modulo π . Put, for $j = 0, 1, 2$,

$$a_j = \cos\left(\frac{2\pi j}{3}\right)e_1 + \sin\left(\frac{2\pi j}{3}\right)e_2,$$

$$b_j = \cos\left(\frac{2\pi j}{3} + \theta\right)e_1 + \sin\left(\frac{2\pi j}{3} + \theta\right)e_2.$$

Then the six rays are pairwise distinct, and

$$\sum_{j=0}^2 P_{a_j} = \sum_{j=0}^2 P_{b_j} = \frac{3}{2} I_V.$$

Thus (a_0, a_1, a_2) and (b_0, b_1, b_2) form a genuine pseudo-context of size $k = 3$ over $\mathbb{R}$.

Proof. In the basis (e_1, e_2) , each projector has the form

$$P_{\cos \varphi e_1 + \sin \varphi e_2} = \begin{pmatrix} \cos^2 \varphi & \cos \varphi \sin \varphi \\ \cos \varphi \sin \varphi & \sin^2 \varphi \end{pmatrix}.$$

For $\varphi_j = 2\pi j/3$ and for $\varphi_j = \theta + 2\pi j/3$, the trigonometric identities

$$\sum_{j=0}^2 \cos(2\varphi_j) = 0, \quad \sum_{j=0}^2 \sin(2\varphi_j) = 0,$$

which hold for any arithmetic progression of three angles spaced $2\pi/3$ apart, imply that the off-diagonal terms cancel while the diagonal terms each sum to $3/2$. Hence both sums equal $\frac{3}{2} I_V$.

The three excluded residue classes modulo π account for all collisions between the two triples:

- $\theta \equiv 0 \pmod{\pi}$: every b_j coincides with a_j or $-a_j$, giving the same ray.
- $\theta \equiv \pi/3 \pmod{\pi}$: the b -triple is a cyclic permutation of the a -triple as a set of rays.
- $\theta \equiv 2\pi/3 \pmod{\pi}$: the b -triple is again a permutation of the a -triple.

For all other values of θ , the six rays are pairwise distinct. Genuineness holds because within each triple the ray angles are separated by $\pi/3$ modulo π , never by $\pi/2$. Hence no two rays within the same triple are orthogonal, so neither triple is a context. $\square$

Remark 23 (Complex trine description). Example 22 has a compact complex analogue. Let $\omega = e^{2\pi i/3}$ and embed V into $\mathbb{C}^n$ as $\text{span}\{e_1, e_2\}$. The *trines*

$$u_j = \frac{1}{\sqrt{2}}(1, \omega^j, 0, \dots, 0), \quad v_j = \frac{1}{\sqrt{2}}(1, e^{i\phi} \omega^j, 0, \dots, 0),$$

with $j = 0, 1, 2$, yield another rank-one decomposition of the same structure operator $\frac{3}{2} I_V$ for any $\phi \in \mathbb{R}$ that is not a multiple of $2\pi/3$. The identity $1 + \omega + \omega^2 = 0$ makes the off-diagonal entries of $\sum_j P_{u_j}$ and $\sum_j P_{v_j}$ cancel, giving the common structure operator $\frac{3}{2} I_V$. This form is most natural in frame-theoretic discussions and in connection with the Weyl–Heisenberg group.

Remark 24 (Minimal ambient dimension). The explicit constructions in Sections IV B, V, and VI all live in the two-dimensional subspace $\text{span}\{e_1, e_2\}$, so they embed into any $\mathbb{C}^n$ or $\mathbb{R}^n$ with $n \geq 2$. This is a statement about the constructions exhibited here, not about all pseudocontexts in all dimensions; indeed, the rotational construction of Section XI produces real pseudocontexts whose rays span all of $\mathbb{R}^n$. No pseudocontext exists in dimension $n = 1$, since there is only one ray.

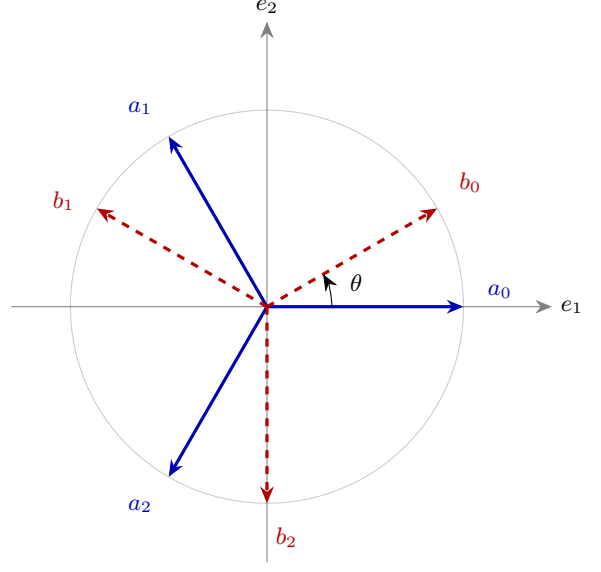


FIG. 1. The Mercedes–Benz pseudocontext of Example 22, drawn for $\theta = \pi/6$. The solid blue rays a_0, a_1, a_2 and the dashed red rays b_0, b_1, b_2 are two distinct unordered triples in the plane V , yet the associated rank-one projectors have the same sum $\frac{3}{2} I_V$.

Remark 25 (Non-scalar structure operators). The trine family is highly symmetric: its structure operator is a scalar multiple of the identity on its support. Non-scalar, equivalently non-tight, examples already occur in the complex two-dimensional construction of Proposition 10; the examples listed here are three-dimensional realizations. Here “non-scalar” means not proportional to the identity on the support; it does not mean that the spectrum is simple. For scalar T the projector-sum statistic is state independent on the support, whereas a non-scalar T ranges over its spectral interval. Explicit three-dimensional implementations of non-scalar structure operators were derived in [3].

- **15-atom hypergraph** (T_1). A pseudocontext of size $k = 3$ yields $\lambda_1 = 2$, $\lambda_2 = \frac{7+\sqrt{21}}{14} \approx 0.827$, $\lambda_3 = \frac{7-\sqrt{21}}{14} \approx 0.173$, with $\text{tr}(T_1) = 3$ and $\lambda_{\max} = 2 > 1$, consistent with Proposition 30.
- **36-atom hypergraph** (T_2). A larger true-implies-false gadget gives $T_2 = \text{diag}(1/5, 1/5, 13/5)$ with $\text{tr}(T_2) = 3$ and $\lambda_{\max} = 13/5 > 1$.

Such examples are useful when one wants to move beyond purely symmetric frame orbits.

VII. SPECTRAL STRUCTURE AND CANONICAL FORMS

In dimension $d = 2$, all spectral properties reduce to the Bloch form of Sec. IV B. We now analyse the general properties of the structure operator $T = \sum_{i=1}^k P_{a_i}$

associated with a k -tuple of unit vectors $(a_1, \dots, a_k)$ in H .

A. Gram representation and spectral equivalence

Definition 26 (Synthesis operator and Gram matrix). Let $A: \mathbb{F}^k \rightarrow H$ be defined by $Ae_i = a_i$, where (e_i) is the standard basis of $\mathbb{F}^k$. The *synthesis operator* is A , its adjoint $A^\dagger: H \rightarrow \mathbb{F}^k$ is the analysis operator, and the *Gram matrix* is

$$G := A^\dagger A = (\langle a_i, a_j \rangle)_{i,j=1}^k.$$

Proposition 27 (Frame operator factorisation). *The structure operator admits the factorisation $T = AA^\dagger$, and $G = A^\dagger A$.*

Proposition 28 (Spectral equivalence). *The nonzero spectra of T and G coincide (including multiplicities):*

$$\sigma(T) \setminus \{0\} = \sigma(G) \setminus \{0\}.$$

In particular, $\text{rank}(T) = \text{rank}(G) \leq \min(k, n)$.

Proof. This is the standard singular-value correspondence between $AA^\dagger$ and $A^\dagger A$: if $AA^\dagger v = \lambda v$ with $\lambda \neq 0$, then $A^\dagger A(A^\dagger v) = \lambda(A^\dagger v)$ and $A^\dagger v \neq 0$. $\square$

B. Spectral constraints

Proposition 29 (Trace identities). *The structure operator satisfies*

$$\text{tr}(T) = k, \quad \text{tr}(T^2) = \sum_{i,j=1}^k |\langle a_i, a_j \rangle|^2.$$

The diagonal terms $i = j$ each contribute 1, giving $\text{tr}(T^2) \geq k$, with equality if and only if the vectors are mutually orthogonal.

Proof. $\text{tr}(T) = \sum_i \text{tr}(P_{a_i}) = k$. For the second identity, $\text{tr}(T^2) = \sum_{i,j} \text{tr}(P_{a_i} P_{a_j}) = \sum_{i,j} |\langle a_i, a_j \rangle|^2$. Diagonal terms contribute k ; off-diagonal terms are non-negative, and vanish simultaneously if and only if all off-diagonal inner products are zero. $\square$

Proposition 30 (Non-orthogonality and maximal eigenvalue). *If the vectors (a_i) are not mutually orthogonal, then $\text{tr}(T^2) > k$ and $\lambda_{\max}(T) > 1$. If, in addition, the nonzero spectrum of T is flat on its support of rank r , then $T = (k/r)I_{\text{supp } T}$ and necessarily $k > r$.*

Proof. The strict trace inequality is Proposition 29. If all eigenvalues of T were at most 1, then $\text{tr}(T^2) \leq \text{tr}(T) = k$, a contradiction. Thus $\lambda_{\max}(T) > 1$. The final assertion follows by writing a flat nonzero spectrum as $k/r, \dots, k/r$. $\square$

Proposition 31 (Majorisation constraint). *Suppose $k \leq n$. Let $\lambda(T) = (\lambda_1, \dots, \lambda_n)$ be the eigenvalues of T in decreasing order (padded with zeros to length n), and let $d = (1, \dots, 1, 0, \dots, 0) \in \mathbb{R}^n$ have k ones and $n - k$ zeros. Then $\lambda(T) \succ d$.*

Proof. The diagonal entries of G are $G_{ii} = \|a_i\|^2 = 1$, so the diagonal of G is $(1, \dots, 1) \in \mathbb{R}^k$. By the Schur–Horn theorem, $\lambda(G) \succ (1, \dots, 1)$. Since G and T share the same nonzero eigenvalues (Proposition 28), padding $\lambda(G)$ with $n - k$ zeros gives $\lambda(T)$, and $\lambda(T) \succ d$ follows.

Overcomplete case $k > n$. When $k > n$ the Gram matrix G is $k \times k$ with rank at most n ; the majorisation is then best stated for the k -dimensional eigenvalue vector of G against $(1, \dots, 1) \in \mathbb{R}^k$, and the Schur–Horn argument applies to G directly. $\square$

Proposition 32 (Operator bounds). *The structure operator satisfies $0 \leq T \leq kI$. Moreover, if $\mu := \max_{i \neq j} |\langle a_i, a_j \rangle|$, then*

$$\lambda_{\max}(T) \leq 1 + (k - 1)\mu.$$

Proof. Positivity $T \geq 0$ is immediate, and $T \leq kI$ since each $P_{a_i} \leq I$. For the eigenvalue bound, apply the Geršgorin circle theorem to G : since $G_{ii} = 1$ and $|G_{ij}| \leq \mu$ for $i \neq j$, every eigenvalue of G lies in a disk centred at 1 with radius at most $(k - 1)\mu$. Since the nonzero eigenvalues of T and G coincide, $\lambda_{\max}(T) \leq 1 + (k - 1)\mu$. $\square$

C. Commutant and symmetry

Definition 33 (Isometric commutant). Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. The isometric commutant of T is

$$\mathcal{C}_{\mathbb{F}}(T) := \{U \in \mathcal{U}_{\mathbb{F}}(H) : UT = TU\},$$

where $\mathcal{U}_{\mathbb{C}}(H) = U(H)$ and $\mathcal{U}_{\mathbb{R}}(H) = O(H)$.

Proposition 34 (Structure of the commutant). *Let*

$$T = \sum_{\alpha} \lambda_{\alpha} P_{\alpha}$$

be the spectral decomposition of T , with distinct eigenvalues λ_{α} and eigenspace dimensions $m_{\alpha} = \text{rank } P_{\alpha}$. Then

$$\mathcal{C}_{\mathbb{C}}(T) \cong \prod_{\alpha} U(m_{\alpha}),$$

whereas

$$\mathcal{C}_{\mathbb{R}}(T) \cong \prod_{\alpha} O(m_{\alpha}).$$

Remark 35. Degeneracy of a *positive* eigenspace enlarges the commutant and can generate continuous families of decompositions. Rotations in the zero eigenspace act trivially on every decomposition, since all frame vectors lie in the support. Over $\mathbb{R}$, a simple positive spectrum

has a discrete commutant on the support; over $\mathbb{C}$ it still has the relative phase group $U(1)^r$. The latter already generates the complex $k = 2$ family. Scalar structure operators maximize the spatial commutant, which explains the special richness of tight frames.

D. Canonical normal form

Proposition 36 (Spectral normal form). *Every structure operator is unitarily (respectively orthogonally) equivalent to $\text{diag}(\lambda_1, \dots, \lambda_r, 0, \dots, 0)$ with $\lambda_i > 0$, $\sum_{i=1}^r \lambda_i = k$, and $r = \text{rank}(T)$.*

E. Schur–Horn characterisation of decompositions

Theorem 37 (Schur–Horn characterisation). *Let T be a positive operator of rank r with spectral decomposition*

$$T = \sum_{i=1}^r \lambda_i |e_i\rangle\langle e_i|, \quad \lambda_1 \geq \dots \geq \lambda_r > 0.$$

Set $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$. A decomposition

$$T = \sum_{j=1}^k P_{a_j}, \quad \|a_j\| = 1,$$

exists only if

$$\sum_{i=1}^r \lambda_i = k$$

and $r \leq k$. Equivalently, it exists if and only if the Schur–Horn condition

$$(\lambda_1, \dots, \lambda_r, 0, \dots, 0) \succ (1, \dots, 1) \in \mathbb{R}^k$$

holds, where the left-hand side is padded with $k - r$ zeros. Given positivity, $r \leq k$, and trace k , this majorisation condition is automatic: a decreasing k -vector with average 1 majorises $(1, \dots, 1)$. Hence a positive T admits such a decomposition if and only if $\text{tr} T = k$ and $\text{rank} T \leq k$.

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. For a fixed T , every such decomposition corresponds to a matrix

$$M \in \mathbb{F}^{r \times k}$$

satisfying

$$MM^\dagger = \Lambda, \quad \|m_j\|^2 = 1, \quad j = 1, \dots, k,$$

where m_j is the j -th column of M , via

$$a_j = \sum_{i=1}^r (m_j)_i e_i.$$

Conversely, every such matrix M defines a unit-norm rank-one decomposition of T . Equivalently, write $M = \Lambda^{1/2} V$ with $VV^\dagger = I_r$ and $\text{diag}(V^\dagger \Lambda V) = \mathbf{1}$.

Proof. The Schur–Horn condition is necessary because the Gram matrix of any such decomposition has diagonal entries all equal to 1 and eigenvalues $(\lambda_1, \dots, \lambda_r, 0, \dots, 0)$. Conversely, by the Schur–Horn theorem over $\mathbb{R}$ and $\mathbb{C}$, if this majorisation condition holds, there exists a positive semidefinite $k \times k$ Gram matrix G with this spectrum and diagonal $(1, \dots, 1)$. Factoring $G = M^\dagger M$, with $M \in \mathbb{F}^{r \times k}$ after restriction to the rank- r support, gives columns of norm 1 and $MM^\dagger = \Lambda$ after choosing the eigenbasis of T . This yields the desired decomposition.

Let $T = AA^\dagger$ with columns a_j . In the eigenbasis of T , write $A = UM$ where U is the unitary embedding the eigenbasis of T on its support into H . Then $MM^\dagger = U^\dagger T U|_{\text{supp}(T)} = \Lambda$, and $\|m_j\| = \|a_j\| = 1$. The converse is a direct computation. $\square$

Corollary 38 (Isometric freedom). *Among all rank-one decompositions $T = \sum_{j=1}^k |a_j\rangle\langle a_j|$ into k not necessarily unit-norm vectors, any two are related by $A \mapsto AU$ for some unitary matrix U (over $\mathbb{C}$, $U \in U(k)$, and over $\mathbb{R}$, $U \in O(k)$). The unit-norm equations select a generally singular subset of this orbit; no uniform codimension statement is needed here. For pseudocontexts the relevant quotient is by independent column phases (or signs) and permutations, since rays—not vectors—are the data.*

F. Characterisation of pseudocontexts

Proposition 39 (Operator equivalence criterion). *Two k -tuples of distinct rays form a pseudocontext if and only if, in the eigenbasis of their common structure operator T , their coefficient matrices M_A, M_B satisfy*

$$M_A M_A^\dagger = M_B M_B^\dagger = \Lambda, \quad \|(m_A)_j\| = \|(m_B)_j\| = 1 \quad \text{for all } j,$$

and the two resulting ray sets are disjoint.

Remark 40. Pseudocontexts correspond to distinct points in the space $\{M \in \mathbb{F}^{r \times k} : MM^\dagger = \Lambda, \|m_j\| = 1\} / \sim$, where $\sim$ identifies decompositions that yield the same set of rays.

G. Extremal spectral bounds

Proposition 41 (Rayleigh bounds). *For any density operator ρ ,*

$$\lambda_{\min}(T) \leq \text{tr}(\rho T) \leq \lambda_{\max}(T).$$

Both bounds are tight, achieved by pure states on the corresponding eigenvectors of T .

VIII. HYPERGRAPH-FORCED AND REPRESENTATION-DEPENDENT IDENTITIES

Definition 42 (Graph-forced identity). Let $\Gamma = (V, \mathcal{E})$ be a finite orthogonality hypergraph, with contexts $E \in$

$\mathcal{E}$. A state on Γ is a function

$$p : V \rightarrow [0, 1]$$

satisfying

$$\sum_{v \in E} p(v) = 1 \quad \text{for every } E \in \mathcal{E}.$$

Given two k -element subsets $A, B \subseteq V$, the identity

$$\sum_{a \in A} p(a) = \sum_{b \in B} p(b)$$

is called graph-forced if it holds for every state p on Γ . A covering proof is a linear certificate of such an identity obtained by subtracting two equal-cardinality multisets of context equations whose auxiliary vertex multiplicities cancel.

The distinction is combinatorial rather than topological. A representation-dependent identity follows from a particular equality of projector sums. A graph-forced identity follows from the normalized context equations alone and therefore holds for every hypergraph state, including every classical two-valued state. Neither property is, by itself, a contextuality witness.

A. Graph-forced pseudocontexts

For certain hypergraphs, a linear combination of context equations forces a pseudocontext identity. The covering method used below is one such certificate: two collections of contexts cover every auxiliary vertex with the same multiplicity while leaving different target ray families. When the hyperedges are represented by complete orthonormal bases, the same identity transfers to quantum probabilities.

B. Representation-dependent pseudocontexts: the $k = 2$ case

For the bare orthogonality hypergraph induced by the four rays in Proposition 10, no context equation contains both members of either nonorthogonal pair. Its hypergraph states can therefore violate the projector-sum identity. This proves only that the *bare* hypergraph does not force the equality; it does not rule out a finite auxiliary extension. The octagon below is an example of how auxiliary contexts can close such loose ends.

IX. THE OCTAGON PSEUDOCONTEXT VIA A COVERING CERTIFICATE

The following 20-vertex, 12-context hypergraph supplies a compact graph-forcing certificate [2].

The hypergraph consists of 20 vertices and 12 contexts of size three. We label the 16 outer vertices along the perimeter as follows: let $v_0, v_1, \dots, v_7$ be the midpoints of the eight octagon edges, and let $v_{01}, v_{12}, \dots, v_{70}$ be the eight corners. The perimeter is formed by eight straight-line contexts:

$$E_i = \{v_{i-1,i}, v_i, v_{i,i+1}\} \quad \text{for } i = 0, \dots, 7 \pmod{8}.$$

Notice that adjacent midpoints, such as v_1 and v_2 , are not on the same context; rather, their respective perimeter contexts E_1 and E_2 intertwine at the shared corner vertex v_{12} .

The remaining four inner contexts connect opposite midpoints and pass through the four inner vertices a_1, a_2 and b_1, b_2 :

$$\begin{aligned} D_0 &= \{v_0, a_1, v_4\}, & D_2 &= \{v_2, a_2, v_6\}, \\ D_1 &= \{v_1, b_2, v_5\}, & D_3 &= \{v_3, b_1, v_7\}. \end{aligned}$$

The graph-forcing certificate below compares two coverings. First, let

$$A = D_0 + D_2 + E_1 + E_3 + E_5 + E_7.$$

Every outer vertex occurs once in A , and a_1, a_2 occur once.

Likewise, let

$$B = \{D_1, D_3, E_0, E_2, E_4, E_6\}.$$

Every outer vertex occurs once in B , and b_1, b_2 occur once. Thus the incidence-vector identity is

$$\chi_A - \chi_B = \chi_{a_1} + \chi_{a_2} - \chi_{b_1} - \chi_{b_2}.$$

Applying a normalized hypergraph state p gives

$$p(a_1) + p(a_2) = p(b_1) + p(b_2).$$

The two coverings are drawn in Fig. 2.

A. Octagon coordinates at $C \neq 1$

In the octagon coordinatization—that is, the search for a faithful orthogonal representation (FOR) [4, 5] of the uniform hypergraph [6] in which vectors assigned to vertices within each hyperedge are required to be mutually orthogonal—it is convenient to parametrize the opening of the inner two-pairs by a single positive parameter C . Since only rays matter, all vectors below are understood up to a nonzero scalar multiple.

Proposition 43 (Meaning of the parameter C). *Let $s > 0$, set $C := s^2$, and choose representatives*

$$a_1 = (s, 1, 0), \quad a_2 = (s, -1, 0),$$

with the companion normal-form pair

$$b_1 = (s, i, 0), \quad b_2 = (s, -i, 0).$$

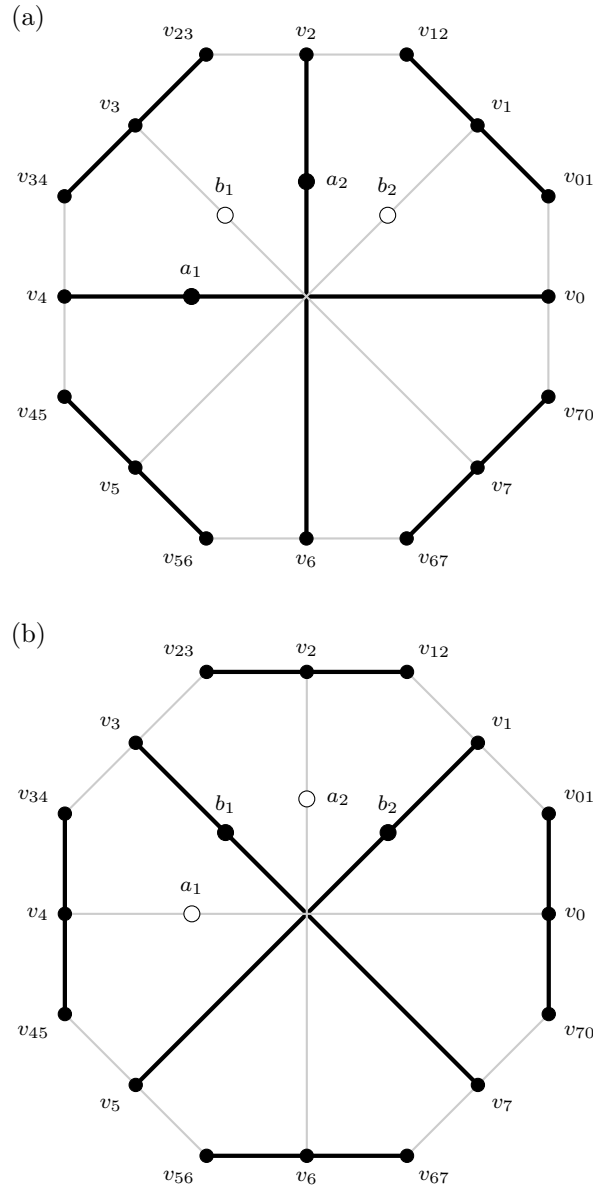


FIG. 2. The two context coverings used in the incidence-vector certificate. Bold lines are selected contexts; filled vertices are covered.

Then the chosen normalized representatives have inner product

$$\langle \hat{a}_1, \hat{a}_2 \rangle = \frac{C-1}{C+1}.$$

Consequently, the invariant principal angle between the rays satisfies

$$\cos \theta_{\text{ray}} = \frac{|C - 1|}{C + 1}.$$

In particular, $C = 1$ is the orthogonal case; C and $1/C$ have the same principal ray angle.

Proof. The normalized representatives of a_1 and a_2 are

$$\frac{(s, 1, 0)}{\sqrt{s^2 + 1}}, \quad \frac{(s, -1, 0)}{\sqrt{s^2 + 1}}.$$

Their inner product equals $(s^2 - 1)/(s^2 + 1)$. Taking its absolute value gives the stated ray-angle formula after substituting $C = s^2$. The same computation applies to the b -pair. $\square$

Remark 44. The parameter C measures the nonorthogonality of the inner two-pairs. In the normal form above the companion pair uses the phases i and $-i$. Later, in the explicit $C \neq 1$ example, this normal-form choice is replaced by a phase-generalized pair $(s, \eta, 0), (s, -\eta, 0)$

with $|\eta| = 1$. This changes the concrete coordinatization but not the common two-projector structure of the inner pseudocontext.

For $x, y \in \mathbb{C}^3$ we use the Hermitian cross product

$$x \boxtimes y := \overline{x \times y},$$

where $x \times y$ is the ordinary bilinear cross product. Then

$$\langle x, x \boxtimes y \rangle = \langle y, x \boxtimes y \rangle = 0$$

with respect to the Hermitian inner product $\langle u, v \rangle = \bar{u}^T v$.

Proposition 45 (Parametrized octagon coordinates). *Let $s > 0$, $C = s^2$, and $N = 1 + C$. Work in the normal form*

$$a_1 = (s, 1, 0), a_2 = (s, -1, 0), b_1 = (s, i, 0), b_2 = (s, -i, 0).$$

Set

$$\omega = i, \quad \mu_j = \omega^{2-j}, \quad j \in \mathbb{Z}/8\mathbb{Z},$$

where the index arithmetic is taken modulo 8 to reflect the closed cyclic perimeter of the octagon. Thus

$$(\mu_0, \dots, \mu_7) = (-1, i, 1, -i, -1, i, 1, -i).$$

Choose $t_0, t_1, t_2, t_3 \in \mathbb{C}^\times$ (the set of non-zero complex numbers), and impose the Hermitian antipodal height relation

$$t_{j+4} = -\frac{N}{\overline{t_j}}, \quad j = 0, 1, 2, 3.$$

The eight midpoint rays are

$$v_j = (1, s\mu_j, t_j), \quad j = 0, \dots, 7.$$

With this choice the four inner triples

$$\{v_0, a_1, v_4\}, \{v_2, a_2, v_6\}, \{v_1, b_2, v_5\}, \{v_3, b_1, v_7\}$$

are automatically contexts.

Define the Gram kernel of the midpoint rays by

$$G_{jk} := \langle v_j, v_k \rangle = 1 + C \overline{\mu_j} \mu_k + \overline{t_j} t_k.$$

The corner rays are obtained by the Hermitian cross-product convention

$$v_{j,j+1} = v_j \boxtimes v_{j+1} := \overline{v_j \times v_{j+1}}, \quad j \in \mathbb{Z}/8\mathbb{Z}.$$

Each such corner is orthogonal to its two adjacent midpoints. Therefore the remaining condition for the eight perimeter triples

$$E_j = \{v_{j-1,j}, v_j, v_{j,j+1}\}, \quad j \in \mathbb{Z}/8\mathbb{Z},$$

to be contexts is equivalently the cyclic Gram system

$$G_{j-1,j} G_{j,j+1} = G_{j-1,j+1} G_{j,j}, \quad j \in \mathbb{Z}/8\mathbb{Z}. \quad (7)$$

In expanded midpoint form this is

$$(1 + C \overline{\mu_{j-1}} \mu_j + \overline{t_{j-1}} t_j)(1 + C \overline{\mu_j} \mu_{j+1} + \overline{t_j} t_{j+1}) - (1 + C \overline{\mu_{j-1}} \mu_{j+1} + \overline{t_{j-1}} t_{j+1})(N + |t_j|^2) = 0. \quad (8)$$

Thus any choice of t_0, t_1, t_2, t_3 satisfying these closure equations yields an orthogonal representation of the octagon hypergraph. It is faithful exactly when the listed rays are pairwise distinct as vertices and no orthogonalities occur except those prescribed by the twelve contexts.

Proof. The antipodal height relation gives, for $j = 0, 1, 2, 3$,

$$\langle v_j, v_{j+4} \rangle = 1 + C \overline{\mu_j} \mu_{j+4} + \overline{t_j} t_{j+4} = 1 + C + \overline{t_j} t_{j+4} = N - N = 0.$$

Moreover, the displayed values of μ_j make v_0, v_4 orthogonal to a_1 , v_2, v_6 orthogonal to a_2 , v_1, v_5 orthogonal to b_2 , and v_3, v_7 orthogonal to b_1 . Hence the four inner triples are contexts.

For the perimeter, the vector $x \boxtimes y$ is orthogonal to both x and y with respect to the Hermitian inner product. Indeed, for

$$v_j = (1, s\omega^{2-j}, t_j),$$

a direct computation gives

$$v_j \boxtimes v_{j+1} = (s\omega^{j-2}(\overline{t_{j+1}} - \omega \overline{t_j}), \overline{t_j} - \overline{t_{j+1}}, s\omega^{j-2}(\omega - 1)),$$

or, up to a nonzero scalar,

$$v_{j,j+1} \propto (s(\overline{t_{j+1}} - \omega \overline{t_j}), \omega^{2-j}(\overline{t_j} - \overline{t_{j+1}}), s(\omega - 1)).$$

Thus $v_{j,j+1}$ is orthogonal to v_j and v_{j+1} . The only remaining orthogonality in the perimeter context E_j is the orthogonality of the two adjacent corners. The standard Gram-determinant identity for the two cross products gives

$$\langle v_{j-1}, v_j \rangle \langle v_j, v_{j+1} \rangle - \langle v_{j-1}, v_{j+1} \rangle \|v_j\|^2 = 0,$$

which is precisely (7); expanding the entries of G gives (8). $\square$

Example 46 (An explicit $C \neq 1$ pseudocontext FOR). Set

$$C = \frac{49 - 5\sqrt{69}}{26}, \quad s = \sqrt{C}, \quad N = 1 + C, \quad r = \sqrt{N}.$$

Then $C > 0$ and $C \neq 1$; indeed C is the smaller positive root of

$$13C^2 - 49C + 13 = 0.$$

Let

$$\eta = \frac{5 + 12i}{13}$$

and

$$\zeta = \frac{14 + 6\sqrt{69}}{65} + i \frac{-21 + 4\sqrt{69}}{65}.$$

Both η and ζ have unit modulus.

Define the two inner pairs

$$a_1 = (s, 1, 0), \quad a_2 = (s, -1, 0),$$

and

$$b_1 = (s, \eta, 0), \quad b_2 = (s, -\eta, 0).$$

The pairs $\{a_1, a_2\}$ and $\{b_1, b_2\}$ are nonorthogonal because

$$\langle a_1, a_2 \rangle = \langle b_1, b_2 \rangle = C - 1 \neq 0.$$

Moreover, since $\eta \notin \{\pm 1\}$, the four rays a_1, a_2, b_1, b_2 are pairwise distinct. Since $|\eta| = 1$, their normalized projector sums agree:

$$P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}.$$

Thus the two inner pairs form a genuine complex pseudocontext.

Now put

$$(\mu_0, \dots, \mu_3) = (-1, \eta, 1, -\eta), \quad (\mu_4, \dots, \mu_7) = (-1, \eta, 1, -\eta), \\ (\tau_0, \dots, \tau_3) = (1, \zeta, -i, -i\zeta), \quad (\tau_4, \dots, \tau_7) = (-1, -\zeta, i, i\zeta),$$

and define the eight midpoint rays by

$$v_j = (1, s\mu_j, r\tau_j), \quad j = 0, \dots, 7.$$

The corner rays are

$$v_{j,j+1} = v_j \boxtimes v_{j+1} := \overline{v_j} \times v_{j+1}, \quad j \in \mathbb{Z}/8\mathbb{Z}.$$

The twelve contexts are the eight perimeter contexts

$$\{v_{j-1,j}, v_j, v_{j,j+1}\}, \quad j \in \mathbb{Z}/8\mathbb{Z},$$

together with the four inner contexts

$$\{v_0, a_1, v_4\}, \quad \{v_2, a_2, v_6\}, \\ \{v_1, b_2, v_5\}, \quad \{v_3, b_1, v_7\}.$$

Together with the diagnostic audit in the following remark, these coordinates give an orthogonal representation of the octagon hypergraph carrying the displayed $C \neq 1$ pseudocontext.

Remark 47 (Reduction of the symmetric ansatz). For the phase-generalized symmetric ansatz in Example 46, the eight perimeter closure equations become identical once the unit-phase condition $|\zeta| = 1$ is imposed. Before imposing this condition, the alternating equations differ only by a multiple of $|\zeta|^2 - 1$. This is best seen before substituting the displayed value of C . Keep

$$\eta = \frac{5 + 12i}{13}, \quad \zeta = x + iy, \quad x, y \in \mathbb{R},$$

and impose the same cyclic pattern

$$(\mu_0, \dots, \mu_3) = (-1, \eta, 1, -\eta), \quad (\mu_4, \dots, \mu_7) = (-1, \eta, 1, -\eta), \\ (\tau_0, \dots, \tau_3) = (1, \zeta, -i, -i\zeta), \quad (\tau_4, \dots, \tau_7) = (-1, -\zeta, i, i\zeta),$$

with midpoint heights $t_j = r\tau_j$, $r^2 = N = 1 + C$. Substitution in (8) gives the same equation for every $j \in \mathbb{Z}/8\mathbb{Z}$. Written in real and imaginary parts, this single complex equation is equivalent to

$$7Cx - 17Cy - 13C - 13x + 13y + 13 = 0$$

and $-7C^2x + 17C^2y + 13C^2 - 20Cx + 30Cy + 2C - 13x + 13y + 13 = 0$. Solving these two linear equations for x and y gives

$$x = \frac{104 - 76C}{65(1 + C)}, \quad y = \frac{39 - 81C}{65(1 + C)}.$$

The unit-circle condition $|\zeta| = 1$, i.e. $x^2 + y^2 = 1$, then reduces to

$$\frac{48(13C^2 - 49C + 13)}{325(1 + C)^2} = 0.$$

Thus the symmetric ansatz closes precisely when

$$13C^2 - 49C + 13 = 0.$$

For the smaller positive root $C = (49 - 5\sqrt{69})/26$, the above formulas for x and y give exactly

$$\zeta = \frac{14 + 6\sqrt{69}}{65} + i \frac{-21 + 4\sqrt{69}}{65},$$

as used in the example.

Remark 48 (Numerical faithfulness audit). For the coordinates in Example 46, a direct Hermitian inner-product check on the full set of twenty rays gives

$$\max_{\{x,y\} \text{ prescribed}} \frac{|\langle x, y \rangle|}{\|x\| \|y\|} < 1.4 \times 10^{-16}.$$

There are 36 prescribed orthogonal pairs among the twelve contexts, and all vanish to this tolerance. Among the remaining 154 unprescribed pairs, the smallest normalized inner product is

$$0.2 = \frac{1}{5},$$

so no unintended orthogonality occurs. All twenty ray representatives are nonzero and projectively distinct; the smallest projective separation between distinct listed rays is

$$0.4472135955 = \frac{1}{\sqrt{5}}.$$

Finally,

$$\|P_{a_1} + P_{a_2} - P_{b_1} - P_{b_2}\|_F < 2.8 \times 10^{-16}.$$

Thus the displayed coordinates pass a strong numerical faithfulness audit. For a formal faithfulness theorem, the submission should also include an exact algebraic or certified-interval certificate that every one of the 154 unprescribed inner products is nonzero; floating-point tolerances alone are not a proof.

B. Exact $\mathbb{C}^3$ coordinates at $C = 1$

At $C = 1$ the inner pairs become orthogonal, so the size-two identity is nongenuine. The octagon nevertheless admits the following closed-form algebraic realization in $\mathbb{C}^3$. The vectors are ray representatives, so normalization is inessential; the configuration is shown in Fig. 3. Appendix B 2 lists 24 separating two-valued states for the resulting 21-vertex, 14-context hypergraph, and hence a partition-logic representation [7–9].

Example 49 (Exact $\mathbb{C}^3$ realization at $C = 1$). Let

$$p = \frac{\sqrt{14} + \sqrt{2}}{4}, \quad q = \frac{\sqrt{14} - \sqrt{2}}{4}.$$

Then

$$p^2 + q^2 = 2, \quad pq = \frac{3}{4}.$$

A convenient choice of the inner rays sharing the additional context vertex $c = (0, 0, 1)$ is

$$\begin{aligned} a_1 &= (1, 1, 0), & a_2 &= (1, -1, 0), \\ b_1 &= (1, i, 0), & b_2 &= (1, -i, 0). \end{aligned}$$

One compatible choice of the eight midpoint rays is

$$\begin{aligned} v_0 &= (1, -1, \sqrt{2}), & v_4 &= (1, -1, -\sqrt{2}), \\ v_1 &= (1, i, p + iq), & v_5 &= (1, i, -p - iq), \\ v_2 &= (1, 1, -i\sqrt{2}), & v_6 &= (1, 1, i\sqrt{2}), \\ v_3 &= (1, -i, q - ip), & v_7 &= (1, -i, -q + ip). \end{aligned}$$

One compatible choice of the eight corner rays is

$$\begin{aligned} v_{01} &= (-p + i(q + \sqrt{2}), \sqrt{2} - p + iq, 1 - i), \\ v_{12} &= (\sqrt{2} - p + iq, p - i(q + \sqrt{2}), 1 + i), \\ v_{23} &= (q + \sqrt{2} + ip, -q + i(\sqrt{2} - p), -1 + i), \\ v_{34} &= (q + i(p - \sqrt{2}), q + \sqrt{2} + ip, -1 - i), \\ v_{45} &= (p - i(q + \sqrt{2}), -\sqrt{2} + p - iq, 1 - i), \\ v_{56} &= (-\sqrt{2} + p - iq, -p + i(q + \sqrt{2}), 1 + i), \\ v_{67} &= (-q - \sqrt{2} - ip, q - i(\sqrt{2} - p), -1 + i), \\ v_{70} &= (-q - i(p - \sqrt{2}), -q - \sqrt{2} - ip, -1 - i). \end{aligned}$$

A direct symbolic check shows that each listed context is orthogonal. The identities $p^2 + q^2 = 2$ and $pq = \frac{3}{4}$ are sufficient for the simplification. Hence the octagon from Section IX admits an exact $\mathbb{C}^3$ realization at $C = 1$.

Remark 50. The coordinates above are not unique. Any common unitary transformation, together with arbitrary rescalings of the individual ray representatives, yields an equivalent realization.

Remark 51 (Phase symmetry). The exact $\mathbb{C}^3$ realization exploits a complex phase symmetry in the third coordinate. This is one of the reasons why the construction is much more rigid over $\mathbb{R}$ than over $\mathbb{C}$.

C. No faithful real realization at $C = 1$

Example 49 provides an exact realization of the octagon hypergraph in $\mathbb{C}^3$ at the orthogonal limit $C = 1$, where the inner pseudocontext splits into two contexts. We show that no faithful realization by real rays exists.

Proposition 52 (No faithful real realization at $C = 1$). *The octagon hypergraph of Section IX, augmented by the two inner $C = 1$ contexts intersecting at the vertex c , admits no faithful representation by rays in $\mathbb{R}^3$, with its three-element contexts represented as orthonormal bases of $\mathbb{R}^3$.*

Proof. Assume a faithful real realization exists. By a global orthogonal transformation, fix $c = (0, 0, 1)^T$. All rays sharing a context with c lie in the xy -plane. Using rotational freedom in that plane, set

$$a_1 = (1, 0, 0)^T, \quad a_2 = (0, 1, 0)^T.$$

Let

$$b_1 = (\cos \theta, \sin \theta, 0)^T, \quad b_2 = (-\sin \theta, \cos \theta, 0)^T,$$

with $\theta \notin \frac{\pi}{2}\mathbb{Z}$ by faithfulness. Define $m = \tan \theta \neq 0$ and $M = 1 + m^2 > 1$.

Let the midpoint rays be $v_j = (X_j, Y_j, Z_j)^T$.

Nonplanarity of midpoint rays. Every midpoint lies in exactly one inner context, paired with one of the planar reference vectors: $v_0, v_4 \in D_0$ (with a_1); $v_2, v_6 \in D_2$ (with a_2); $v_1, v_5 \in D_1$ (with b_2); $v_3, v_7 \in D_3$ (with b_1). Suppose $Z_j = 0$, so that v_j lies in the xy -plane. If v_j is paired with a_1 (resp. a_2), orthogonality forces $X_j = 0$ (resp. $Y_j = 0$), leaving v_j proportional to a_2 (resp. a_1). If v_j is paired with b_1 or b_2 , the relations $Y = mX$ or $X = -mY$ together with $Z_j = 0$ force coincidence with one of the b -rays. In every case this violates faithfulness. Hence $Z_j \neq 0$ for all j , and we may normalize

$$v_j = (X_j, Y_j, 1)^T.$$

The inner contexts impose:

$$\begin{aligned} D_0 : & \quad X_0 = X_4 = 0, \quad Y_0 Y_4 + 1 = 0, \\ D_1 : & \quad Y_1 = mX_1, \quad Y_5 = mX_5, \quad X_1 X_5 M + 1 = 0, \\ D_3 : & \quad X_3 = -mY_3, \quad X_7 = -mY_7, \quad Y_3 Y_7 M + 1 = 0. \end{aligned}$$

Thus

$$X_5 = -\frac{1}{MX_1}, \quad Y_7 = -\frac{1}{MY_3}.$$

The relations

$$X_1 X_5 M + 1 = 0, \quad Y_3 Y_7 M + 1 = 0$$

immediately imply

$$X_1, X_5, Y_3, Y_7 \neq 0,$$

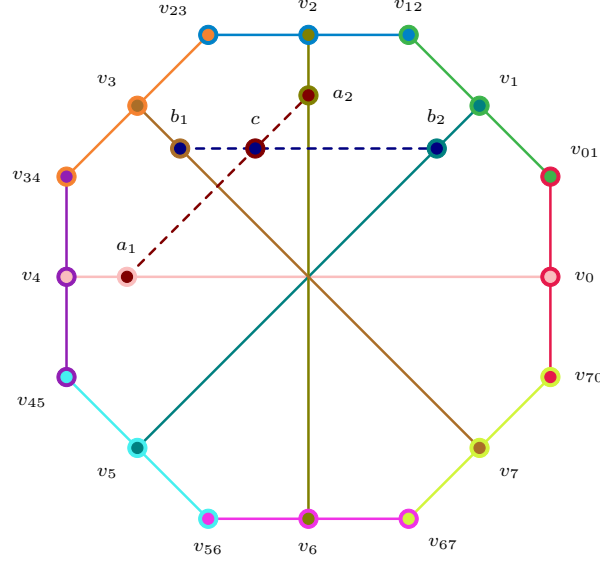


FIG. 3. The $C = 1$ augmentation of the octagon in Fig. 2. The 12 original contexts are drawn with solid lines in 12 distinct colors. The two additional $C = 1$ contexts are dashed and meet at the intertwining vertex c . Concentric disks mark vertices incident with two contexts.

since $M = 1 + m^2 > 0$.

For perimeter contexts, the condition that adjacent corner rays are orthogonal yields

$$(v_{j-1} \cdot v_j)(v_j \cdot v_{j+1}) - (v_{j-1} \cdot v_{j+1})\|v_j\|^2 = 0.$$

Evaluating at $j = 0$ and $j = 4$ gives the perimeter closure equations. At $j = 0$, the perimeter closure equation factors as

$$Y_0((mX_1Y_7 - 1)Y_0 + Y_7 + mX_1) = 0.$$

The branch $Y_0 = 0$ is impossible because the inner context D_0 gives

$$Y_0Y_4 + 1 = 0.$$

Therefore the nonzero branch yields the displayed formula for Y_0 , and similarly for Y_4 :

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1Y_7}, \quad Y_4 = \frac{Y_3 + mX_5}{1 - mY_3X_5}, \quad (9)$$

where the denominators are nonvanishing: if $1 - mX_1Y_7 = 0$, the closure equation at $j = 0$ reduces to $Y_7 + mX_1 = 0$, and combining this with $mX_1Y_7 = 1$ yields $-m^2X_1^2 = 1$, which has no real solution. The case $1 - mY_3X_5 = 0$ at $j = 4$ is excluded analogously.

Substituting the inner relations into (9) and simplifying the compound fractions yields

$$Y_0 = \frac{mMX_1Y_3 - 1}{MY_3 + mX_1}, \quad Y_4 = \frac{MX_1Y_3 - m}{MX_1 + mY_3}.$$

Using $Y_0Y_4 = -1$ from D_0 , set $u = X_1Y_3$. Expanding the product of these fractions, collecting terms in u , and applying the identity $M = 1 + m^2$ causes all mixed terms to

cancel, yielding the master constraint (see Appendix A)

$$(M^2u^2 + mu + 1) + M(X_1^2 + Y_3^2) = 0. \quad (10)$$

Since $X_1, Y_3 \in \mathbb{R}$,

$$X_1^2 + Y_3^2 \geq 2|X_1Y_3| = 2|u|,$$

and therefore the left-hand side is bounded below by

$$f(u) = M^2u^2 + mu + 1 + 2M|u|.$$

Using $M = 1 + m^2$, one has $2M - |m| > 0$, hence

$$M^2u^2 + mu + 2M|u| \geq M^2u^2 + |u|(2M - |m|) \geq 0,$$

with equality only at $u = 0$. Consequently $f(u) \geq 1$ for all $u \in \mathbb{R}$, contradicting the required vanishing. Therefore no real realization exists. $\square$

Remark 53 (Reduced closure constraints). The contradiction uses the added contexts through c , the three inner contexts D_0, D_1, D_3 , and the two reduced perimeter closure equations at $j = 0$ and $j = 4$. Each closure equation itself is derived after eliminating corner rays from three perimeter contexts. Thus this is a useful reduced algebraic subsystem, not a five-context subhypergraph obstruction.

Remark 54. The result is a dimension-three separation under a specific completeness assumption: Proposition 52 rules out a faithful real ray representation in $\mathbb{R}^3$ when every three-element context is an orthonormal basis. It does not rule out a higher-dimensional real representation in which the listed triples are merely orthogonal triples.

X. PHYSICAL INTERPRETATION AND CONSTRAINTS

A pseudocontext equates two state-independent *Born-weight sums*: the rank-one projectors within either family need not commute and are therefore normally estimated on separately prepared copies. For $A_i = P_{a_i}$ and $B_i = P_{b_i}$, the two linear statistics

$$\rho \mapsto \sum_i \text{tr}(\rho A_i), \quad \rho \mapsto \sum_i \text{tr}(\rho B_i)$$

agree on the whole state space. They need not be probabilities: for a genuine pseudocontext their common operator T has $\lambda_{\max}(T) > 1$.

There is nevertheless a direct POVM realization. Since $0 \leq T \leq kI$,

$$E_i = \frac{1}{k}P_{a_i}, \quad E_0 = I - \frac{1}{k}T, \quad F_i = \frac{1}{k}P_{b_i}, \quad F_0 = I - \frac{1}{k}T$$

define two POVMs with the same coarse-grained effect T/k . This is the appropriate operational sense in which the two families are equivalent.

The spectral decomposition of T establishes absolute quantum limits on these probability sums. By Proposition 41,

$$\lambda_{\min}(T) \leq \sum_{i=1}^k p_\rho(a_i) \leq \lambda_{\max}(T). \quad (11)$$

These bounds are tight.

For the 36-atom hypergraph example [3], the structure operator $T_2 = \text{diag}(1/5, 1/5, 13/5)$ confines the quantum probability sum to $[1/5, 13/5] = [0.2, 2.6]$. A classical hidden-variable model based on two-valued $(\{0, 1\})$ truth assignments to projections—subject to the consistency requirement that exactly one atom in each orthogonal basis receives the value 1—would allow the sum to range over $\{0, 1, 2, 3\}$, which includes the extreme values 0 and 3. Because T_2 has no eigenvalue equal to 0 or 3, quantum mechanics forbids these extremes; the quantum system cannot reach them. This compares the quantum-realizable and hypergraph-classical state spaces, but it is not by itself a noncontextuality-inequality violation.

This state-independent additive equivalence is related to, but distinct from, the operational notion of contextuality introduced by Spekkens [10], which applies to arbitrary experimental procedures (preparations, transformations, and unsharp measurements) rather than to sharp projective measurements alone. Accordingly, pseudocontextuality, graph forcing, and Kochen–Specker noncolorability are distinct notions.

In the language of quantum information, $(A_1, \dots, A_k)$ and $(B_1, \dots, B_k)$ are distinct rank-one decompositions of the same positive operator. When this operator is proportional to the identity on its support, the families form tight frames; after normalisation they yield POVMs.

XI. GENUINELY THREE-DIMENSIONAL REAL PSEUDOCONTEXTS

The constructions of Sections V–VI produce real pseudocontexts whose rays are confined to a two-dimensional subspace. We now exhibit the full-support tight-frame mechanism. It exploits the spectral structure developed in Section VII and yields continuous families of real pseudocontexts whose rays span all of $\mathbb{R}^n$. The starting point is the observation that when the structure operator T is proportional to the identity, the orthogonal commutant (Proposition 34) is the full orthogonal group, so every spatial rotation produces a new rank-one decomposition of the same operator.

A. The rotation principle for tight frames

Proposition 55 (Rotation principle). *Let $(a_1, \dots, a_k)$ be a unit-norm tight frame in $\mathbb{R}^n$ with pairwise distinct rays, i.e.,*

$$T = \sum_{i=1}^k P_{a_i} = \frac{k}{n} I_n.$$

For every $R \in O(n)$, the rotated tuple $(Ra_1, \dots, Ra_k)$ is again a unit-norm tight frame with the same structure operator T . The collision set

$$S^* = \{R \in O(n) : [Ra_j] = [a_i] \text{ for some } i, j\}$$

is a closed proper subvariety of $O(n)$ of codimension at least $n - 1$, and hence has Haar measure zero. For every $R \in O(n) \setminus S^$ the $2k$ rays $[a_1], \dots, [a_k], [Ra_1], \dots, [Ra_k]$ are pairwise distinct, and the tuples $(a_1, \dots, a_k)$ and $(Ra_1, \dots, Ra_k)$ form a pseudocontext of size k , genuine whenever $k > n$.*

Proof. Conjugation by $R \in O(n)$ fixes any scalar multiple of the identity, so

$$RTR^T = T.$$

Since $P_{Ra_i} = R P_{a_i} R^T$, summing gives

$$\sum_{i=1}^k P_{Ra_i} = R \left(\sum_{i=1}^k P_{a_i} \right) R^T = RTR^T = T,$$

proving that the rotated tuple is a unit-norm tight frame with the same structure operator.

For each ordered pair (i, j) and each sign $\varepsilon \in \{\pm 1\}$, the locus $\{R \in O(n) : Ra_j = \varepsilon a_i\}$ is a single coset of the stabiliser $\text{Stab}_{O(n)}(a_j) \cong O(n - 1)$, hence a closed submanifold of codimension $n - 1$ in $O(n)$. The collision set S^* is the finite union of these submanifolds, hence a closed subvariety of codimension at least $n - 1$ and Haar measure zero.

For $R \in O(n) \setminus S^*$, no Ra_j is collinear with any a_i . Since the original rays are pairwise distinct and rotations preserve ray distinctness, the $2k$ rays are pairwise

distinct. When $k > n$, no k -tuple of unit vectors in $\mathbb{R}^n$ can be orthogonal, so neither tuple is a context, and the pseudocontext is genuine. $\square$

Remark 56. The genericity condition $R \in O(n) \setminus S^*$ is mild: S^* has Haar measure zero in $O(n)$, so a Haar-random rotation produces a pseudocontext with probability one. The valid parameter space $O(n) \setminus S^*$ is open and dense, and itself a smooth manifold of dimension $\binom{n}{2}$.

Remark 57 (Spectral interpretation). Proposition 55 is the maximally degenerate case of the isometric commutant analysis of Section VII: when the spectrum of T collapses to a single eigenvalue, the commutant $\mathcal{C}_{\mathbb{R}}(T)$ becomes the full orthogonal group, and every rotation gives a fresh rank-one decomposition. This sharpens the informal observation made in Remark 21 for the two-dimensional support of the trine: tight frames maximize this spatial commutant symmetry. They are not the only structure operators with continuous decomposition families.

Corollary 58 (Spectrum of tight pseudocontexts). *Let $n \geq 2$. A genuine pseudocontext with structure operator $T = \lambda I_n$ exists if and only if λ is of the form*

$$\lambda = \frac{k}{n}$$

for some integer $k > n$.

Proof. If $T = \lambda I_n$ arises from a k -tuple of unit vectors, then by Proposition 29,

$$\text{tr}(T) = k.$$

Since $\text{tr}(\lambda I_n) = n\lambda$, it follows that $\lambda = k/n$. Here k is the cardinality of the tuple, hence an integer. Thus, for fixed n , scalar tight structure operators can occur only for

$$\lambda \in \left\{ \frac{k}{n} : k \in \mathbb{N} \right\}.$$

Genuineness forces $k > n$, because $k = n$ would make the frame an orthonormal basis by Proposition 2.

Conversely, for every integer $k > n$, the standard finite unit-norm tight-frame existence theorem supplies a frame $(a_1, \dots, a_k)$ with pairwise distinct rays in $\mathbb{R}^n$ (and hence in $\mathbb{C}^n$) satisfying

$$\sum_{i=1}^k P_{a_i} = \frac{k}{n} I_n.$$

Proposition 55 then implies that, for a generic rotation $R \in O(n) \setminus S^*$, the tuple $(Ra_1, \dots, Ra_k)$ yields a distinct decomposition of the same operator. The two tuples therefore form a genuine pseudocontext. $\square$

B. The tetrahedral pseudocontext

The simplest nontrivial application of Proposition 55 in genuine three-dimensional ambient space is provided by the regular tetrahedron, the unique four-ray unit-norm tight frame in $\mathbb{R}^3$ up to orthogonal equivalence and relabeling (or, for vectors, also independent signs).

Example 59 (Tetrahedral pseudocontext). Let

$$\begin{aligned} a_1 &= (0, 0, 1), \\ a_2 &= \left(\frac{2\sqrt{2}}{3}, 0, -\frac{1}{3}\right), \\ a_3 &= \left(-\frac{\sqrt{2}}{3}, \frac{\sqrt{6}}{3}, -\frac{1}{3}\right), \\ a_4 &= \left(-\frac{\sqrt{2}}{3}, -\frac{\sqrt{6}}{3}, -\frac{1}{3}\right) \end{aligned}$$

be the vertices of a regular tetrahedron inscribed in the unit sphere of $\mathbb{R}^3$, satisfying $\langle a_i, a_j \rangle = -1/3$ for $i \neq j$. Choose any $R \in SO(3) \setminus S^*$ (i.e. R is not a rotation that maps any a_j onto an $\pm a_i$), and define

$$b_j := Ra_j, \quad j = 1, \dots, 4.$$

Then the eight rays $[a_1], \dots, [a_4], [b_1], \dots, [b_4]$ are pairwise distinct, and

$$\sum_{j=1}^4 P_{a_j} = \sum_{j=1}^4 P_{b_j} = \frac{4}{3} I_3,$$

so $(a_1, \dots, a_4)$ and $(b_1, \dots, b_4)$ form a genuine pseudocontext of size $k = 4$ in $\mathbb{R}^3$ whose eight rays span all of $\mathbb{R}^3$.

A concrete instance is provided by the quarter-turn about the x -axis,

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \in SO(3),$$

which yields

$$\begin{aligned} b_1 &= (0, -1, 0), \\ b_2 &= \left(\frac{2\sqrt{2}}{3}, \frac{1}{3}, 0\right), \\ b_3 &= \left(-\frac{\sqrt{2}}{3}, \frac{1}{3}, \frac{\sqrt{6}}{3}\right), \\ b_4 &= \left(-\frac{\sqrt{2}}{3}, \frac{1}{3}, -\frac{\sqrt{6}}{3}\right). \end{aligned}$$

A direct check shows that none of the b_j is collinear with any a_i , confirming that the eight rays are distinct.

Proof. The matrix $A = [a_1 \ a_2 \ a_3 \ a_4]$ satisfies $AA^T = \frac{4}{3} I_3$ by direct computation, so $T = \frac{4}{3} I_3$. Since $R \in O(3)$,

$$\sum_{j=1}^4 P_{b_j} = R \left(\sum_{j=1}^4 P_{a_j} \right) R^T = R \cdot \frac{4}{3} I_3 \cdot R^T = \frac{4}{3} I_3,$$

giving the common structure operator. The hypothesis $R \in SO(3) \setminus S^*$ excludes ray collisions, so the eight

rays are pairwise distinct. Within the a -tuple, the inner products $\langle a_i, a_j \rangle = -1/3 \neq 0$ for $i \neq j$ show that no two distinct vectors are orthogonal, so the tuple is not a context; the b -tuple is its image under the orthogonal map R and inherits this property. The pseudocontext is therefore genuine, in agreement with Proposition 55 specialised to $n = 3$, $k = 4$. $\square$

Remark 60 (Orbit of a fixed tetrahedron). For the fixed tetrahedral frame in Example 59, the family obtained by varying $R \in SO(3) \setminus S^*$ is parametrized by an open dense subset of $SO(3)$, modulo the finite tetrahedral symmetry group. This is a 3-dimensional orbit of one fixed frame, not the full moduli space of all size-4 tight frames in $\mathbb{R}^3$.

Remark 61 (Fixed-frame versus full moduli). The tetrahedral example exhibits one particularly symmetric point in the space of size-4 tight frames in $\mathbb{R}^3$. The point of the example is only that this one orbit already yields a genuine pseudocontext spanning all of $\mathbb{R}^3$.

Remark 62 (Physical interpretation). The physical tetrahedral POVM has effects $E_j = \frac{3}{4}P_{a_j}$, which sum to I_3 ; the rotated frame gives another such POVM. Equality of their total probabilities is therefore normalization. The example is useful as a robust projector-sum identity, but it is not by itself a contextuality effect.

Conjecture 63 (No finite graph-forcing extension for the generic tetrahedral pseudocontext). *Let $(a_1, \dots, a_4)$ and $(b_1, \dots, b_4) = (Ra_1, \dots, Ra_4)$ be the tetrahedral pseudocontext of Example 59, with $R \in SO(3) \setminus S^*$ outside the exceptional algebraic loci where additional ray coincidences or unintended orthogonalities occur. Then, generically in R , there is no finite orthogonality hypergraph Γ over $\mathbb{R}^3$, generated by adding finitely many auxiliary rays and contexts to these eight rays, whose state space forces*

$$\sum_{j=1}^4 p(a_j) = \sum_{j=1}^4 p(b_j).$$

Equivalently, the tetrahedral pseudocontext is generically representation-dependent rather than graph-forced.

Remark 64 (Heuristic motivation). The absence of orthogonalities among the eight displayed rays makes a covering construction non-obvious. This is heuristic motivation only: it does not exclude other finite auxiliary incidence patterns and is not evidence sufficient to prove the conjecture.

Remark 65 (Complex analogue). The corresponding question over $\mathbb{C}$ remains open. A naive parameter count is inconclusive: after quotienting the central phase, the effective unitary parameter space has eight real dimensions, and one must also account for the possible auxiliary incidence patterns.

C. The tightness obstruction at $k = 3$

The rotation principle of Proposition 55 cannot be applied at $k = 3$ in $\mathbb{R}^3$, because tightness in this regime

collapses to orthogonality.

Proposition 66. *Let (a_1, a_2, a_3) be a unit-norm tight frame in $\mathbb{R}^3$, i.e. $T = I_3$. Then (a_1, a_2, a_3) is an orthonormal basis, hence a context.*

Proof. By Proposition 29, $\text{tr}(T^2) = k + \sum_{i \neq j} |\langle a_i, a_j \rangle|^2$. With $T = I_3$ and $k = 3$ this gives $3 = 3 + \sum_{i \neq j} |\langle a_i, a_j \rangle|^2$, forcing all off-diagonal Gram entries to vanish. $\square$

Corollary 67. *Every genuine real pseudocontext of size $k = 3$ whose rays span all of $\mathbb{R}^3$ has non-scalar structure operator, hence is not tight on its support, and therefore lies outside the rotational orbit construction of Proposition 55.*

Such genuinely three-dimensional, non-scalar $k = 3$ pseudocontexts do exist: explicit coordinatisations were obtained in [3] (recalled in Remark 25), with structure operators having spectra $\{2, (7 + \sqrt{21})/14, (7 - \sqrt{21})/14\}$ and $\{1/5, 1/5, 13/5\}$ respectively. These configurations are produced not by acting on a single tight base configuration with a continuous symmetry, but by solving the canonical fibre conditions of Theorem 37 directly with a prescribed non-scalar structure operator, the off-diagonal Gram data being constrained by an underlying orthogonality hypergraph rather than recoverable from a continuous group orbit.

D. Larger sizes

For every $k \geq 4$, unit-norm tight frames in $\mathbb{R}^3$ exist in abundance, and Proposition 55 produces a 3-parameter family of size- k pseudocontexts whenever the chosen frame has finite ray-stabiliser. Among the most symmetric examples we have the Platonic frames and their pyramid generalisations:

- $k = 4$: regular tetrahedron, $T = \frac{4}{3}I_3$ (Example 59);
- $k = 5$: a regular square pyramid whose four base vectors are elevated to angle $\arccos(1/\sqrt{6})$ from the apex, $T = \frac{5}{3}I_3$;
- $k = 6$: one representative from each of the six antipodal vertex pairs of a regular icosahedron, $T = 2I_3$;
- $k = 7$: a regular hexagonal pyramid whose six base vectors are elevated to angle $\arccos(\sqrt{2}/3)$ from the apex, $T = \frac{7}{3}I_3$;
- $k = 12$: a union of three generically rotated tetrahedral tight frames, chosen with no ray coincidences, gives $T = 4I_3$.

Regular tetrahedral and icosahedral ray frames give real ETFs for $k = 4$ and $k = 6$. Gerzon's bound excludes real ETFs with $k > 6$ in $\mathbb{R}^3$. The nonexistence of the $k = 5$ ETF is a separate fact (for example, it follows

from Naimark complementation), and should not be attributed to Gerzon's bound. Absence of an ETF does not mean absence of other highly symmetric tight frames.

However, to compensate for the loss of perfect symmetry, these non-ETF frames acquire internal degrees of freedom. The expected local parameter count for the unit-norm tight-frame variety in $\mathbb{R}^3$, modulo rotations and away from singular points, is $2k - 8$. Thus, while $k = 4$ represents a rigid shape, the highly symmetric “square pyramid” of $k = 5$ sits inside a positive-dimensional space of tight-frame deformations (e.g., stretching into rectangular or rhombic pyramids), and the $k = 7$ configuration has still more deformation directions. Each point in such a configuration space generates its own 3-parameter rotational family of real pseudocontexts spanning $\mathbb{R}^3$.

Remark 68. Together with the planar Mercedes-Benz construction (Example 22) and the non-scalar $k = 3$ realisations in $\mathbb{R}^3$ of [3] (cf. Remark 25), Example 59 completes a coherent picture of real pseudocontexts in low ambient dimension: every $k \geq 3$ admits genuine real realizations spanning $\mathbb{R}^3$. Those for $k = 3$ are necessarily non-scalar (Corollary 67), those for $k = 4$ and $k = 6$ admit highly rigid, rotationally parametrised ETF families, and those for $k = 5$ and $k \geq 7$ need not be equiangular. Their tight-frame spaces can nevertheless have continuous internal deformations. We do not make a separate numerical-continuation claim here without reproducible coordinates and code.

Example 69 (Pyramidal tight frames for $k = 5$ and $k = 7$). The $k = 5$ and $k = 7$ cases need not be equiangular, but highly symmetric unit-norm tight frames can still be constructed via regular pyramids. For $k = 5$, a tight frame with $T = \frac{5}{3}I_3$ is formed by a square pyramid with apex $v_1 = (0, 0, 1)$ and four base vectors at an angle of $\arccos(1/\sqrt{6})$ from the apex:

$$v_{2,3} = \left(\pm \sqrt{\frac{5}{6}}, 0, \frac{1}{\sqrt{6}} \right), \quad v_{4,5} = \left(0, \pm \sqrt{\frac{5}{6}}, \frac{1}{\sqrt{6}} \right).$$

Similarly, for $k = 7$, a tight frame with $T = \frac{7}{3}I_3$ is given by a hexagonal pyramid with apex $v_1 = (0, 0, 1)$ and six base vectors at an angle of $\arccos(\sqrt{2}/3)$ from the apex:

$$v_{j+2} = \left(\frac{\sqrt{7}}{3} \cos \frac{j\pi}{3}, \frac{\sqrt{7}}{3} \sin \frac{j\pi}{3}, \frac{\sqrt{2}}{3} \right) \quad \text{for } j \in \{0, \dots, 5\}.$$

Unlike the rigid ETFs at $k = 4$ and $k = 6$, the expected local parameter count for the unit-norm tight-frame equations in $\mathbb{R}^3$ is $2k - 8$ after quotienting by rotations. Thus these $k = 5$ and $k = 7$ pyramids should be viewed as highly symmetric points inside positive-dimensional spaces of tight realisations, rather than as isolated Platonic configurations.

XII. OPEN PROBLEMS

Open Problem 70 (Classification over $\mathbb{C}^n$). Classify, up to unitary equivalence, all pseudocontexts of size k in

$\mathbb{C}^n$. Even for small (k, n) , the orbit structure is not yet transparent.

Open Problem 71 (Lattice representability). Characterise which orthomodular lattices carrying a pseudocontext-type additive identification admit a faithful embedding into some $L(\mathbb{C}^n)$, sharpening the partial results of [3, 11].

Open Problem 72 (Frame-symmetry characterisation in the partially degenerate case). For a positive operator T with spectral decomposition

$$T = \sum_{\alpha} \lambda_{\alpha} P_{\alpha},$$

describe the decomposition space

$$\mathcal{D}(T) = \left\{ (a_1, \dots, a_k) : \sum_i P_{a_i} = T, \|a_i\| = 1 \right\}$$

modulo column phases, permutations, and the commutant

$$\prod_{\alpha} U(\text{rank } P_{\alpha}).$$

In particular, determine the commutant-orbit strata and the residual moduli of inequivalent decompositions for partially degenerate spectra.

Open Problem 73 (Hypergraph-forced higher-dimensional pseudocontexts). The rotation construction of Section XI produces continuous families of operator-level pseudocontexts spanning all of $\mathbb{R}^n$. For generic tetrahedral examples, it is unknown whether a finite auxiliary orthogonality hypergraph can force the identity. In contrast, in [3, 11] and in Section IX of the present paper, the orthogonality relations themselves induce condition (iii) of Definition 3 for every representation in which each listed hyperedge is a complete orthonormal basis. The systematic classification of hypergraphs that force a pseudocontext identity, and of the higher-dimensional configurations they admit, remains open. This The rotation construction and the graph-forced constructions may overlap; the tetrahedral case is a concrete unresolved test case.

Open Problem 74 (Classification of minimal real pseudocontexts). Theorem 13(ii) and the proposition that follows it (Proposition 15) establish that genuine real pseudocontexts of size $k = 2$ do not exist in any dimension, while Example 22 demonstrates their existence for $k = 3$ in two-dimensional subspaces and [3] exhibits $\mathbb{R}^3$ -spanning instances. Thus $k = 3$ is the minimal size for genuine real pseudocontexts. Classify all such minimal ($k = 3$) examples up to orthogonal equivalence over $\mathbb{R}$, distinguishing the two-dimensional case (where the doubled-angle projective-circle description reduces the problem to equal-centroid decompositions of three unit vectors, with scalar structure operators only in the zero-centroid tight-frame subcase) from the genuinely three-dimensional case (where, by Corollary 67, the structure operator is necessarily non-scalar).

Open Problem 75 (Moduli of tight real pseudocontexts). For each $k \geq n + 1$, describe the moduli space $\mathcal{M}_{k,n}^{\text{tight}} = \{\text{unit-norm tight frames of size } k \text{ in } \mathbb{R}^n\} / O(n) \times ((\mathbb{Z}_2)^k \rtimes S_k)$, where independent signs and S_k do not change the ray frame. A pseudocontext is a pair of such ray frames with no cross-collisions; Proposition 55 describes only its congruent rotational sub-stratum. The case $(k, n) = (4, 3)$ reduces to a ray-level point (the regular tetrahedron), while $(k, n) = (5, 3)$ already has positive dimension.

XIII. CONCLUSION

Pseudocontexts are equal rank-one projector decompositions, and their minimum genuine size depends sharply on the scalar field: $k = 2$ over $\mathbb{C}$, but $k = 3$ over $\mathbb{R}$. This is an operator-level fact about quantum Born-rule states. A separate question is whether an equality is forced by normalized context equations. The octagon gives a covering certificate of the latter kind, whereas the tight-frame and tetrahedral examples are representation-dependent constructions. These distinctions should be retained when connecting the subject to contextuality: a projector-sum identity is neither automatically a contextuality witness nor incompatible with a separating family of two-valued states.

DATA AVAILABILITY

The exact coordinates appear in the text. The scripts and complete two-valued-state tables used to verify the numerical audits and the enumerations are available from the corresponding authors upon reasonable request.

ACKNOWLEDGMENTS

We are grateful to Josef Tkadlec for providing a *Pascal* program that computes and analyses the set of two-valued states of collections of contexts.

This text was partially created and revised with assistance from large language models. All content, ideas, and prompts were provided by the authors.

This research was funded in part by the Austrian Science Fund (FWF) 10.55776/PIN5424624 and the Czech Science Foundation (GAČR) 25-20013L. The authors declare no conflict of interest.

Appendix A: Derivation of the Master Constraint

We provide a step-by-step verification of the algebraic derivation leading from the perimeter closure conditions to the master constraint (10) used in the proof of Proposition 52.

The starting point is the reduced perimeter closure condition stated in the proof. The three midpoint rays v_{j-1}, v_j, v_{j+1} do not form a context; rather, orthogonality of the two corner rays obtained from their adjacent pairs yields

$$(v_{j-1} \cdot v_j)(v_j \cdot v_{j+1}) - (v_{j-1} \cdot v_{j+1})\|v_j\|^2 = 0. \quad (\text{A1})$$

Evaluating (A1) at $j = 0$ and $j = 4$ produces the relations

$$Y_0 = \frac{Y_7 + mX_1}{1 - mX_1Y_7}, \quad (\text{A2})$$

$$Y_4 = \frac{Y_3 + mX_5}{1 - mY_3X_5}, \quad (\text{A3})$$

as stated in the proof. The inner contexts impose the constraints

$$Y_7 = -\frac{1}{MY_3}, \quad X_5 = -\frac{1}{MX_1}, \quad M = 1 + m^2, \quad (\text{A4})$$

together with the requirement $Y_0Y_4 = -1$ from context D_0 .

Simplifying the fractions

Substituting Y_7 from (A4) into (A2):

$$Y_0 = \frac{-\frac{1}{MY_3} + mX_1}{1 - mX_1(-\frac{1}{MY_3})} = \frac{mMX_1Y_3 - 1}{MY_3 + mX_1}. \quad (\text{A5})$$

Substituting X_5 from (A4) into (A3):

$$Y_4 = \frac{Y_3 + m(-\frac{1}{MX_1})}{1 - mY_3(-\frac{1}{MX_1})} = \frac{MX_1Y_3 - m}{MX_1 + mY_3}. \quad (\text{A6})$$

Constructing the product condition

Let $u = X_1Y_3$. Using the constraint $Y_0Y_4 = -1$ from context D_0 together with (A5) and (A6):

$$\frac{mMu - 1}{mX_1 + MY_3} \cdot \frac{Mu - m}{MX_1 + mY_3} = -1. \quad (\text{A7})$$

Cross-multiplying (A7) yields

$$(mMu - 1)(Mu - m) = -(mX_1 + MY_3)(MX_1 + mY_3). \quad (\text{A8})$$

Expanding both sides

The left-hand side of (A8) expands as

$$\begin{aligned} \text{LHS} &= mM^2u^2 - m^2Mu - Mu + m \\ &= mM^2u^2 - Mu(m^2 + 1) + m. \end{aligned} \quad (\text{A9})$$

Since $M = 1 + m^2$, the coefficient $m^2 + 1$ equals M , giving

$$\text{LHS} = mM^2u^2 - M^2u + m. \quad (\text{A10})$$

The right-hand side of (A8) expands as

$$\begin{aligned} \text{RHS} &= -[mMX_1^2 + m^2X_1Y_3 + M^2X_1Y_3 + mMY_3^2] \\ &= -[u(M^2 + m^2) + mM(X_1^2 + Y_3^2)]. \end{aligned} \quad (\text{A11})$$

Cancellation and the master constraint

Equating (A10) and (A11):

$$mM^2u^2 - M^2u + m = -u(M^2 + m^2) - mM(X_1^2 + Y_3^2). \quad (\text{A12})$$

Adding M^2u to both sides, the $-M^2u$ term cancels:

$$mM^2u^2 + m = -m^2u - mM(X_1^2 + Y_3^2). \quad (\text{A13})$$

Bringing all terms to the left and dividing by $m \neq 0$ (since $\theta \notin \frac{\pi}{2}\mathbb{Z}$ by faithfulness) yields the master constraint (10) in the proof of Proposition 52.

Positivity contradiction

Since $X_1, Y_3 \in \mathbb{R}$, the inequality $X_1^2 + Y_3^2 \geq 2|X_1Y_3| = 2|u|$ holds. The left-hand side of (10) therefore satisfies

$$\begin{aligned} &M^2u^2 + mu + 1 + M(X_1^2 + Y_3^2) \\ &\geq M^2u^2 + mu + 1 + 2M|u| \\ &\geq M^2u^2 + |u|(2M - |m|) + 1. \end{aligned} \quad (\text{A14})$$

With $M = 1 + m^2 > 1$, we have $2M - |m| = 2 + 2m^2 - |m| > 0$ for all real m . Consequently the expression in (A14) is strictly positive for every $u \in \mathbb{R}$, contradicting (10). This completes the algebraic verification of the master constraint and the resulting impossibility of a real realization.

Appendix B: Set representation of the octagon hypergraph via two-valued states

As described in Section 2.3 of Ref. [12], any finite atomic logic equipped with a separating set of dispersion-free probability measures (two-valued states) can be faithfully represented as a partition logic [7–9]. In this formulation, the set representation is constructed by “inverting” the set of two-valued states: every vertex v (atom) of the hypergraph is mapped to the set of indices of those two-valued states that evaluate to 1 (“true”) on it. Explicitly, if M denotes the complete set of two-valued states s_k , the set representation of a vertex v is given by

$$p(v) = \{k \mid s_k(v) = 1, k \in M\}. \quad (\text{B1})$$

Under this mapping, every context (orthogonal basis) translates into a disjoint partition of the total state space M .

1. Representation of the incomplete configuration

An exhaustive enumeration of the two-valued states on the tight pseudocontext configuration introduced in Section IX (the incomplete octagon hypergraph depicted in Fig. 2) reveals exactly 26 such states. Because this set of 26 states mutually separates every pair of vertices, the resulting set representation is faithful.

Let us index the set of two-valued states as $M_{26} = \{1, 2, \dots, 26\}$. The full evaluation grid—referred to here as the Travis matrix [8, 13]—is presented in Table II, where the states s_k act as the rows and the vertices as the columns. By taking the vertical union of the evaluated indices where $s_k(v) = 1$, we obtain the subsets representing each vertex. Traversing the 16 perimeter vertices in counterclockwise order starting from v_{56} , followed by the 4 inner vertices a_1, b_1, a_2, b_2 , the explicit mapping is as follows:

$$\begin{aligned} p(v_{56}) &= \{1, 2, 3, 4, 5, 6, 7, 8, 9\}, \\ p(v_6) &= \{10, 11, 12, 13, 14, 15, 16, 17\}, \\ p(v_{67}) &= \{18, 19, 20, 21, 22, 23, 24, 25, 26\}, \\ p(v_7) &= \{1, 2, 3, 4, 10, 11, 12, 13\}, \\ p(v_{70}) &= \{5, 6, 7, 8, 9, 14, 15, 16, 17\}, \\ p(v_0) &= \{1, 2, 10, 11, 18, 19, 20, 21\}, \\ p(v_{01}) &= \{3, 4, 12, 13, 22, 23, 24, 25, 26\}, \\ p(v_1) &= \{1, 5, 6, 7, 10, 14, 18, 19\}, \\ p(v_{12}) &= \{2, 8, 9, 11, 15, 16, 17, 20, 21\}, \\ p(v_2) &= \{1, 3, 5, 6, 18, 22, 23, 24\}, \\ p(v_{23}) &= \{4, 7, 10, 12, 13, 14, 19, 25, 26\}, \\ p(v_3) &= \{5, 8, 15, 16, 18, 20, 22, 23\}, \\ p(v_{34}) &= \{1, 2, 3, 6, 9, 11, 17, 21, 24\}, \\ p(v_4) &= \{4, 5, 7, 8, 12, 15, 22, 25\}, \\ p(v_{45}) &= \{10, 13, 14, 16, 18, 19, 20, 23, 26\}, \\ p(v_5) &= \{11, 12, 15, 17, 21, 22, 24, 25\}, \\ p(a_1) &= \{3, 6, 9, 13, 14, 16, 17, 23, 24, 26\}, \\ p(b_1) &= \{6, 7, 9, 14, 17, 19, 21, 24, 25, 26\}, \\ p(a_2) &= \{2, 4, 7, 8, 9, 19, 20, 21, 25, 26\}, \\ p(b_2) &= \{2, 3, 4, 8, 9, 13, 16, 20, 23, 26\}. \end{aligned}$$

2. Representation of the full configuration

When considering the full context configuration of Fig. 3, which includes the additional intertwining point c and two more contexts ($\{a_1, c, a_2\}$ and $\{b_1, c, b_2\}$), the tighter hypergraph imposes stronger constraints and yields a set of exactly 24 two-valued states.

Let us index this new state space as $M_{24} = \{1, 2, \dots, 24\}$. The Travis matrix has dimensions 24×21 ,

TABLE II. The Travis matrix of the 26 two-valued states for the incomplete octagon configuration. Rows $s_1, \dots, s_{26}$ denote the states, and columns represent the vertices.

State	v_{56}	v_6	v_{67}	v_7	v_{70}	v_0	v_{01}	v_1	v_{12}	v_2	v_{23}	v_3	v_{34}	v_4	v_{45}	v_5	a_1	b_1	a_2	b_2
s_1	1	0	0	1	0	1	0	1	0	1	0	0	1	0	0	0	0	0	0	0
s_2	1	0	0	1	0	1	0	0	1	0	0	0	1	0	0	0	0	0	1	1
s_3	1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	1
s_4	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0	1	1
s_5	1	0	0	0	1	0	0	1	0	1	0	1	0	1	0	0	0	0	0	0
s_6	1	0	0	0	1	0	0	1	0	1	0	0	1	0	0	0	1	1	0	0
s_7	1	0	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0
s_8	1	0	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0	0	1	1
s_9	1	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	1	1	1
s_{10}	0	1	0	1	0	1	0	1	0	0	1	0	0	0	1	0	0	0	0	0
s_{11}	0	1	0	1	0	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0
s_{12}	0	1	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0	0
s_{13}	0	1	0	1	0	0	1	0	0	0	1	0	0	0	1	0	1	0	0	1
s_{14}	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	0	1	1	0	0
s_{15}	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	1	0	0	0	0
s_{16}	0	1	0	0	1	0	0	0	1	0	0	1	0	0	1	0	1	0	0	1
s_{17}	0	1	0	0	1	0	0	0	1	0	0	0	1	0	0	1	1	1	0	0
s_{18}	0	0	1	0	0	1	0	1	0	1	0	1	0	0	1	0	0	0	0	0
s_{19}	0	0	1	0	0	1	0	1	0	0	1	0	0	0	1	0	0	1	1	0
s_{20}	0	0	1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	1
s_{21}	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0	1	1	0
s_{22}	0	0	1	0	0	0	1	0	0	1	0	1	0	1	0	1	0	0	0	0
s_{23}	0	0	1	0	0	0	1	0	0	1	0	1	0	0	1	0	1	0	0	1
s_{24}	0	0	1	0	0	0	1	0	0	1	0	0	1	0	0	1	1	1	0	0
s_{25}	0	0	1	0	0	0	1	0	0	0	1	0	0	1	0	1	0	1	1	0
s_{26}	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	0	1	1	1	1

corresponding to the 24 states and the 20 previous vertices plus the new vertex c (see Table III). Extracting the non-zero indices column by column provides the explicit set representation for the full hypergraph:

$$\begin{aligned}
p(v_{56}) &= \{1, 2, 3, 4, 5, 6, 7, 8\}, \\
p(v_6) &= \{9, 10, 11, 12, 13, 14, 15, 16\}, \\
p(v_{67}) &= \{17, 18, 19, 20, 21, 22, 23, 24\}, \\
p(v_7) &= \{1, 2, 3, 4, 9, 10, 11, 12\}, \\
p(v_{70}) &= \{5, 6, 7, 8, 13, 14, 15, 16\}, \\
p(v_0) &= \{1, 2, 9, 10, 17, 18, 19, 20\}, \\
p(v_{01}) &= \{3, 4, 11, 12, 21, 22, 23, 24\}, \\
p(v_1) &= \{1, 5, 6, 7, 9, 13, 17, 18\}, \\
p(v_{12}) &= \{2, 8, 10, 14, 15, 16, 19, 20\}, \\
p(v_2) &= \{1, 3, 5, 6, 17, 21, 22, 23\}, \\
p(v_{23}) &= \{4, 7, 9, 11, 12, 13, 18, 24\}, \\
p(v_3) &= \{5, 8, 14, 15, 17, 19, 21, 22\}, \\
p(v_{34}) &= \{1, 2, 3, 6, 10, 16, 20, 23\}, \\
p(v_4) &= \{4, 5, 7, 8, 11, 14, 21, 24\}, \\
p(v_{45}) &= \{9, 12, 13, 15, 17, 18, 19, 22\}, \\
p(v_5) &= \{10, 11, 14, 16, 20, 21, 23, 24\}, \\
p(a_1) &= \{3, 6, 12, 13, 15, 16, 22, 23\}, \\
p(b_1) &= \{6, 7, 13, 16, 18, 20, 23, 24\}, \\
p(a_2) &= \{2, 4, 7, 8, 18, 19, 20, 24\}, \\
p(b_2) &= \{2, 3, 4, 8, 12, 15, 19, 22\}, \\
p(c) &= \{1, 5, 9, 10, 11, 14, 17, 21\}.
\end{aligned}$$

It is straightforward to verify that every context, in-

cluding the newly added ones, maps to a valid disjoint partition of the complete state space M_{24} . For example, the new context $\{a_1, c, a_2\}$ yields the exact partition $p(a_1) \cup p(c) \cup p(a_2) = M_{24}$. This explicit equivalence confirms that the logical skeleton of the tight pseudocontext is fully robust, supporting a classical, mutually separating set representation of automaton logic alongside its quantum representations.

3. Comparison of deterministic, classical, and quantum probability sums

The indicator or characteristic function $\mathbf{1}_A(x) = 1$ if $x \in A$ and 0 if $x \notin A$. For a vertex v , write

$$s_i(v) = \mathbf{1}_{p(v)}(i),$$

where $p(v) = \{i : s_i(v) = 1\}$ is its support in the Travis representation.

The equality

$$p(a_1) \cup p(a_2) = p(b_1) \cup p(b_2)$$

therefore says only that the two pairs have the same support for the corresponding logical OR. It does not, by itself, imply equality of the arithmetic sums of truth values. Indeed, for two Boolean variables x, y ,

$$x + y = (x \vee y) + (x \wedge y).$$

Equivalently, for indicator (characteristic) functions,

$$\mathbf{1}_A + \mathbf{1}_B = \mathbf{1}_{A \cup B} + \mathbf{1}_{A \cap B}.$$

Thus equality of unions must be supplemented by equality of intersections.

In the 26-state representation one finds

$$p(a_1) \cap p(a_2) = p(b_1) \cap p(b_2) = \{9, 26\},$$

whereas in the 24-state representation one finds

$$p(a_1) \cap p(a_2) = p(b_1) \cap p(b_2) = \emptyset.$$

Together with the corresponding union equalities, this implies that, in these two particular configurations,

$$s_i(a_1) + s_i(a_2) = s_i(b_1) + s_i(b_2)$$

for every two-valued state s_i in the complete enumeration. This pointwise identity should be understood as a property verified from the full Travis matrix, or derived from the context-covering equations of Section IX, not as a consequence of union equality alone.

For deterministic two-valued states, the four-term sum

$$s_i(a_1) + s_i(a_2) + s_i(b_1) + s_i(b_2) = 2[s_i(a_1) + s_i(a_2)]$$

therefore takes the values 0, 2, 4 in the incomplete case, but only 0, 2 in the full $C = 1$ configuration. For classical

hidden-variable mixtures over the enumerated two-valued states,

$$P_{\text{cl}}(v) = \sum_i \lambda_i s_i(v), \quad \lambda_i \geq 0, \quad \sum_i \lambda_i = 1,$$

the four-term sum is no longer restricted to these discrete values, but ranges over their convex hull: $[0, 4]$ in the incomplete case and $[0, 2]$ in the full $C = 1$ case.

On the quantum side, the corresponding equality is expressed by the structure operator

$$T = P_{a_1} + P_{a_2} = P_{b_1} + P_{b_2}.$$

Hence

$$P_Q(a_1) + P_Q(a_2) + P_Q(b_1) + P_Q(b_2) = 2 \operatorname{tr}(\rho T).$$

For a two-dimensional support V , the $k = 2$ Bloch representation gives

$$T = I_V + \frac{1}{2} \vec{R} \cdot \vec{\sigma},$$

with eigenvalues

$$\lambda_{\pm} = 1 \pm \frac{1}{2} \|\vec{R}\|.$$

Thus the maximal quantum four-term sum is

$$2\lambda_{\max}(T) = 2 + \|\vec{R}\|.$$

In the incomplete nonorthogonal case, $\vec{R} \neq 0$, so this maximum is larger than 2, but still strictly below the deterministic classical maximum 4 unless the two rays collapse. In the full $C = 1$ case, orthogonality forces $\vec{R} = 0$, so

$$T = I_V.$$

Since $\{a_1, c, a_2\}$ is a context in the full $C = 1$ configuration, $P_{a_1} + P_c + P_{a_2} = I$, giving $T = I - P_c$. Consequently,

$$0 \leq 2 \operatorname{tr}(\rho T) \leq 2,$$

with equality 2 for states supported entirely on $V = c^{\perp}$. Thus the added contexts remove the deterministic classical excess 4 and bring the classical convex bound for the four-term sum into agreement with the quantum spectral bound.

4. Faithfulness and the Kochen–Specker distinction

The faithfulness of the two set representations above is established directly by the complete Travis matrices: distinct vertices have distinct support sets $p(v)$, and every context is represented as a disjoint partition of the relevant state space M_{26} or M_{24} . Hence these particular hypergraphs admit faithful partition-logic representations.

TABLE III. The Travis matrix of the 24 two-valued states for the full octagon configuration. Rows $s_1, \dots, s_{24}$ denote the states, and columns represent the vertices.

State	v_{56}	v_6	v_{67}	v_7	v_{70}	v_0	v_{01}	v_1	v_{12}	v_2	v_{23}	v_3	v_{34}	v_4	v_{45}	v_5	a_1	b_1	a_2	b_2	c
s_1	1	0	0	1	0	1	0	1	0	1	0	0	1	0	0	0	0	0	0	0	1
s_2	1	0	0	1	0	1	0	0	1	0	0	0	1	0	0	0	0	0	1	1	0
s_3	1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0
s_4	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0	1	1	0
s_5	1	0	0	0	1	0	0	1	0	1	0	1	0	1	0	0	0	0	0	0	1
s_6	1	0	0	0	1	0	0	1	0	1	0	0	1	0	0	0	1	1	0	0	0
s_7	1	0	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0	0
s_8	1	0	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0	0	1	1	0
s_9	0	1	0	1	0	1	0	1	0	0	1	0	0	0	1	0	0	0	0	0	1
s_{10}	0	1	0	1	0	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0	1
s_{11}	0	1	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	0	0	0	1
s_{12}	0	1	0	1	0	0	1	0	0	0	1	0	0	0	1	0	1	0	0	1	0
s_{13}	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	0	1	1	0	0	0
s_{14}	0	1	0	0	1	0	0	0	1	0	0	1	0	1	0	1	0	0	0	0	1
s_{15}	0	1	0	0	1	0	0	0	1	0	0	1	0	0	1	0	0	1	0	0	1
s_{16}	0	1	0	0	1	0	0	0	1	0	0	0	1	0	0	1	1	1	0	0	0
s_{17}	0	0	1	0	0	1	0	1	0	1	0	1	0	0	1	0	0	0	0	0	1
s_{18}	0	0	1	0	0	1	0	1	0	0	1	0	0	0	1	0	0	1	1	0	0
s_{19}	0	0	1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	0	1	1	0
s_{20}	0	0	1	0	0	1	0	0	1	0	0	0	1	0	0	1	0	1	1	0	0
s_{21}	0	0	1	0	0	0	1	0	0	1	0	1	0	1	0	1	0	0	0	0	1
s_{22}	0	0	1	0	0	0	1	0	0	1	0	1	0	0	1	0	1	0	0	1	0
s_{23}	0	0	1	0	0	0	1	0	0	1	0	0	1	0	0	1	1	1	0	0	0
s_{24}	0	0	1	0	0	0	1	0	0	0	1	0	0	1	0	1	0	1	1	0	0

This property should not be understood as automatic for arbitrary finite orthogonality hypergraphs. Kochen–Specker configurations provide the opposite phenomenon: the orthogonality constraints may be so strong that no separating family of globally consistent

two-valued states exists, and in some cases no two-valued state exists at all. Thus the present octagon configurations lie on the classically representable side of the Zierler–Schlessinger/Greechie criterion, whereas Kochen–Specker sets lie on the nonrepresentable side.

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